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From Text to Returns: Financial News Embeddings as Predictors of Short-Term Abnormal Stock Price Movements

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Declaration

The work contained in this thesis is my own unless otherwise stated.

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Abstract

Large language model (LLM) embeddings provide a numerical representation of financial news that can be used to study its relationship with stock-price movements. This thesis uses LLM embeddings of financial news to investigate whether the information contained in news articles can predict the magnitude of subsequent short-term abnormal price movements, and examines the role of firm-specific information in this relationship. Using news articles covering 14 large-cap U.S. stocks from January 2024 to July 2025, the study compares the predictive performance of original "raw news" embeddings with residualised "pure news" embeddings, from which information associated with observable firm characteristics has been removed. Predictive performance is evaluated across 1-day, 5-day, and 21-day forecasting horizons using absolute abnormal returns as the measure of stock-price movement. At the 1-day and 5-day horizons, "raw news" consistently outperforms "pure news", while predictive performance is substantially weaker for both representations at the 21-day horizon, where "pure news" performs better in relative terms. Further analysis shows a positive association between the amount of firm-specific information encoded in an embedding and its predictive performance at the shorter horizons. Moreover, reintroducing firm identity into the "pure news" representation largely restores the predictive performance of "raw news" at the 1-day and 5-day horizons. These findings suggest that firm-specific information is an important component of the information in financial news that is useful for identifying the magnitude of short-term abnormal stock-price movements.

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Chapter 1

Introduction

Financial markets continuously respond to new information about firms and their operating environments. This information can arise from a variety of sources and take many different forms, ranging from developments concerning a firm's operations and financial performance to changes in its competitive, regulatory, or broader economic environment. As investors process such information, their expectations about firms may change, which can in turn affect asset prices. Understanding how information contained in these developments is reflected in prices is therefore a central question in financial economics. A substantial amount of this information is communicated through financial text, particularly through news articles and other forms of publicly available reporting. Unlike structured financial variables, textual information is not readily expressed as a small set of numerical measures. Extracting economically meaningful information from financial text therefore presents a challenge for quantitative analysis.

The increasing availability of large language models (LLMs) provides a new way of studying this problem. Traditional approaches to financial text analysis have often relied on predefined dictionaries or manually constructed textual features. Such approaches reduce complex articles to a relatively small set of measures and may therefore fail to capture information contained in the broader context of the text. LLMs provide an alternative by converting text into high-dimensional numerical representations, commonly referred to as embeddings. Rather than assigning a single sentiment score to an article, an embedding represents the article as a vector whose dimensions collectively encode information about its semantic content. These representations can subsequently be incorporated into conventional quantitative models, allowing information contained in unstructured financial text to be analysed alongside financial data.

Recent research has demonstrated that LLM-generated representations of financial news can contain information relevant to asset pricing and return prediction. For example, LLM-based representations have been used to predict stock returns and construct trading signals [2]. However, the information contained in a news article need not be entirely novel. News about a firm may also reflect persistent characteristics of that firm that were already observable before the article was published. This distinction is important when interpreting the information content of news. If an embedding contains both information specific to the particular news event and information associated with the identity or existing characteristics of the firm, predictive models may exploit both sources of information without distinguishing between them.

This issue is central to the framework developed by [4], whose analysis of the inefficient pricing of news motivates the methodology adopted in this study. The basic intuition of their approach can be expressed simply as:

$$\text{Information in a news article} = \text{Firm-related information} + \text{Remaining information}.$$

The firm-related information in the news representation can be explained by observable firm characteristics, and the remainder represents unexplained information, which [4] refers to as "pure news". Separating these components provides a way of asking more precisely what part of a news representation is associated with subsequent movements in asset prices. The present study is inspired by this framework, but adapts the residualisation procedure to the limitations of the available data.

The objective of this study is therefore to compare how well the information contained in a news article predicts subsequent stock-price movements with how well the information remaining after accounting for observable firm characteristics predicts those movements. Stock-price movements are measured through absolute abnormal returns, which capture the magnitude of a stock's movement irrespective of its direction, after accounting for the broader market movement.

To investigate this problem, the study first represents financial news articles using LLM-generated document embeddings. The resulting high-dimensional representations are then examined in relation to observable firm characteristics. A residualisation framework is developed to remove the component of the embedding that can be attributed to persistent firm-level information, producing a representation referred to throughout the thesis as "pure news". The methodology is adapted from the framework of [4]. However, the approach is modified to suit the structure of the cross-section used in this study, which also leads to a different interpretation of the resulting "pure news" representation. The thesis therefore examines in greater detail what information this residual representation captures and how it can be interpreted in the context of the present study. Following this, the study compares the original news article representation, referred to as "raw news", with the "pure news" representation in terms of their ability to explain variation in absolute abnormal returns. Finally, the study investigates the role of firm identity in these representations to explore possible explanations for differences in their predictive performance for absolute abnormal returns.

The remainder of the thesis is organised as follows. Chapter 2 provides a brief background on LLM embeddings and their applications in financial text analysis. Chapter 3 describes the dataset and the regression framework adapted from [4] to construct the "pure news" representation and examines how this representation can be interpreted. Chapter 4 compares the predictive performance of the "raw news" and "pure news" representations across different forecasting horizons, followed by an investigation into the role of firm identity in their predictive performance. Chapter 5 further elaborates on the results, discusses their limitations, and considers possible directions for future research.

Chapter 2

Background

2.1 Large Language Models

2.1.1 Overview

A language model is a computational system that learns the statistical structure of language by predicting the likelihood of sequences of words or tokens. In its simplest form, a language model predicts the next word given a sequence of previous words. Large language models (LLMs) extend this idea by using neural networks with substantially larger numbers of parameters and training them on large-scale text corpora. A corpus (plural: corpora) refers to a collection of text documents used for training or analysing language models. Through training on diverse corpora, LLMs learn complex patterns in language, including syntactic relationships, semantic associations, and contextual dependencies [9, Chapter 7, pp. 146–148].

Compared with traditional natural language processing approaches, which often rely on manually designed features such as word counts or predefined linguistic rules, LLMs learn numerical representations of language directly from data. This allows them to capture richer semantic relationships and generalise across a wider range of language tasks.

The development of modern LLMs was enabled by the Transformer architecture introduced by [27]. Unlike earlier sequence models that processed text sequentially, Transformers use attention mechanisms to allow each token representation to incorporate information from other tokens in the sequence. This ability to model long-range dependencies forms the foundation of contemporary LLMs.

2.1.2 From Text to Numerical Representations

Tokenisation

LLMs cannot process raw text directly because neural networks operate on numerical inputs rather than character strings. Thus, text must first be converted into a sequence of discrete units known as tokens using a process referred to as tokenisation. Tokenisation provides a standardised representation of text that ensures that different models process the same input consistently. A token may represent a character, a whole word, or punctuation. Modern language models commonly employ subword tokenisation. In subword tokenisation, words are decomposed into smaller, frequently occurring units known as subwords. This approach provides an effective balance between vocabulary size and the ability to represent rare or previously unseen words. Two widely used subword tokenisation algorithms are Byte Pair Encoding (BPE) and the Unigram Language Model [9, Chapter 2, p. 13].

Embeddings

Once text has been converted into tokens, language models then require a numerical representation of these discrete units. A general representation would be any mapping that transforms an object, such as a word, token, or sentence, into a form that a machine learning model can process. These representations often take the form of numerical vectors. An embedding is a specific type of vector representation learned from data such that, in the geometry of the resulting vector space, semantic and syntactic relationships between tokens are captured [9, Chapter 5, pp. 96, 100].

Early approaches to representing words used counts of their surrounding words. A common example is a one-hot representation. In this representation, each word is assigned a vector with a single non-zero entry corresponding to its position in the vocabulary. Although such representations are useful for identification, they do not capture relationships between words. Moreover, because vocabulary sizes can be very large, these vectors become high-dimensional and sparse, with most entries equal to zero. Efficient storage is therefore required when working with such representations [9, Chapter 5, pp. 102–103].

Embeddings address these limitations by representing words as low-dimensional dense vectors. Instead of having one dimension for every vocabulary item, an embedding maps each word or token into a continuous vector space with a much smaller number of dimensions, typically ranging from tens to thousands of dimensions. Unlike sparse representations, embedding vectors contain real-valued entries. The individual dimensions of an embedding usually do not correspond to explicit linguistic concepts, but semantic information is distributed across the entire vector. The underlying assumption is based on the hypothesis that words occurring in similar contexts tend to have similar meanings, and therefore should have similar vector representations [9, Chapter 5, p. 105].

One influential approach for learning embeddings is the Word2Vec framework introduced in [15, 16]. The Skip-Gram model with Negative Sampling (SGNS), which is an architecture in the Word2Vec framework, learns word representations by exploiting the surrounding context of words in large text corpora. Instead of explicitly counting word co-occurrences, the model is trained on a prediction task in which it predicts which surrounding words are likely to appear in its context, given a target word. Words occurring close to the target word are treated as positive examples, while the remaining words are used as negative examples. The learned parameters of the model from this prediction task form the embedding vectors [9, Chapter 5, p. 106].

A key property of embeddings is that relationships between concepts can emerge from their relative positions in the vector space. For example, Word2Vec demonstrated that certain semantic relationships can be represented through vector arithmetic. The well-known example

$$\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$$

shows that the difference between word vectors can encode relational information, such as gender or other semantic transformations [17]. Similar observations have been made for other relationships like geographical relations and grammatical properties. This suggests that embeddings capture aspects of the underlying structure of language.

But traditional embeddings such as Word2Vec are static embeddings, i.e., each word type in the vocabulary is assigned a single vector regardless of the context in which it appears. This creates limitations for words with multiple meanings. For example, the word "bank" has different meanings depending on whether it appears in a financial or geographical context. Modern language models overcome this limitation through contextual embeddings. Contextual embedding models generate representations that depend on the surrounding sequence of tokens instead of assigning one fixed vector to each word. Thus, the same word can receive different vector representations depending on its usage in a sentence [9, Chapter 9, pp. 106, 207].

LLMs generate contextual representations by processing sequences of token embeddings through a neural network architecture called a Transformer. The resulting embeddings encode information about the input text that can be used for downstream tasks requiring semantic understanding. Unlike traditional word embeddings, contextual embeddings are not merely properties of individual words but representations of specific instances of words or complete text sequences within their context [27, 3].

For completeness, a detailed description of the Transformer architecture is provided in Appendix A.

2.1.3 Document Embeddings

Many natural language processing tasks require a representation of an entire text sequence rather than individual token representations. While Transformer models produce contextual representations for each token in the input, these token-level representations can be combined to obtain a single fixed-dimensional vector representing the entire document. Such vector representations are referred to as document embeddings or sentence embeddings, depending on the length of the input text.

A document embedding is designed to capture the overall semantic content of a text while preserving relationships between documents in the embedding space. Documents with similar meanings are therefore expected to have similar vector representations. Several approaches have been proposed to obtain document embeddings from Transformer models. Common methods include using the representation of a designated classification token, averaging the contextual representations of all tokens, or training models specifically to produce high-quality sentence and document embeddings [3, 21].

Recent embedding models extend this idea by training Transformer-based architectures specifically for representation learning, enabling them to produce dense vector representations that are well suited for downstream machine learning tasks [18]. In this study, document embeddings are used to represent financial news articles as fixed-dimensional vectors that serve as the input features for the subsequent quantitative analysis.

2.2 Large Language Models for Financial Text Analysis

Financial markets generate large volumes of unstructured textual information through news articles, company filings, earnings calls, analyst reports, and social media. Extracting economically relevant information from these sources has long been an important research topic, as such information can influence investor expectations and asset prices. Thus, a substantial body of literature has investigated methods for converting financial text into quantitative signals for sentiment analysis, return prediction, portfolio optimisation, risk management, among other applications. The recent development of LLMs has significantly advanced this area by enabling richer semantic understanding of financial text.

Prior to the widespread adoption of LLMs, research on financial text was primarily through lexicon-based and traditional machine learning methods. Lexicon-based approaches rely on predefined dictionaries that assign sentiment scores to words or phrases. These simple methods have been shown to be effective in various applications. For example, [28] showed that domain-specific lexicons applied to financial social media can extract sentiment measures correlated with market trends. Machine learning approaches have also been widely applied to financial text analysis. Using a large StockTwits dataset, [22] demonstrated that feature engineering techniques, such as incorporating bigrams and emojis, can improve the sentiment classification performance of a wide range of machine learning algorithms, while [24] found that lexicon-based approaches can remain competitive with traditional machine learning methods. However, these approaches have several

limitations. Since sentiment often depends on the surrounding context, methods based on predefined dictionaries may fail to correctly interpret the intended meaning of an expression. On top of that, they often struggle with more complex linguistic structures like sarcasm and irony, and may require specialised resources adapted to specific domains [11].

LLMs address many of the limitations of the earlier approaches. Their ability to incorporate context allows them to better interpret complex financial language, including domain-specific terminology, informal expressions, and other forms of unstructured text. They also have the ability to process long documents, enabling the analysis of detailed financial reports, earnings transcripts, and other extensive sources of financial information. Their stronger contextual understanding can also improve robustness against intentionally manipulated text [11].

As such, LLMs have been applied to a wide range of financial tasks using textual data. For instance, [25] employed BERT to extract semantic and sentiment information from financial text on Twitter to improve covariance estimation and portfolio optimisation. Similarly, [12] used LLMs to extract key events from news articles, which were then used as input features for stock return prediction. Their results showed that these structured event representations were effective for cross-sectional stock return prediction while also providing interpretable event-based explanations. While these studies used LLMs to extract information from text, more recent work has begun to use LLM embeddings directly as model inputs. For example, [1] integrated LLM-generated news embeddings with financial time series data to improve Conditional Value-at-Risk (CoVaR) estimation, outperforming models based on traditional sentiment measures. Embeddings generated by models such as BERT, RoBERTa, LLaMA and ChatGPT were used to predict cross-sectional stock returns in [2]. The embeddings were used as fixed features in a linear ridge regression model to forecast one-day-ahead stock returns. Out-of-sample predictive performance was evaluated using the cross-sectional correlation between predicted and realised returns, and these predictions were then used to form daily long-short quintile portfolios. Their results showed that portfolios formed on these signals achieved high Sharpe ratios, for example, 4.78 for ChatGPT, and were robust to transaction costs.

However, the results from applying LLMs to financial news may be affected by the fact that a substantial portion of news content does not represent genuinely new information in a statistical sense. Instead, much of the information conveyed in news articles can be explained by pre-existing firm-level economic characteristics. Motivated by this observation, [4] investigated whether financial markets fully incorporate the information contained in news articles into asset prices. Rather than using the embeddings directly as predictive features, the authors first examined whether the embeddings themselves were predictable from a firm's observable characteristics, such as size, value, profitability, and momentum. To do so, a large set of firm characteristics was regressed on each coordinate of the news embeddings. The residual components of the embedding coordinates were then used to form a new embedding, which is defined as "pure news" and represents information that cannot be explained by a firm's existing economic profile. The return predictability of the original news embeddings was then compared with that of the residualised embeddings. The results showed that removing the predictable component more than doubled the predictive performance of the news signal, with the out-of-sample Sharpe ratio increasing from 1.1 using raw news to 3.1 using "pure news". The authors concluded that the information most relevant for asset pricing is the genuinely novel component of financial news rather than the information already reflected in firm characteristics.

Despite the promising results of LLM-based approaches, their performance should be interpreted with caution due to the potential for look-ahead bias. Since modern LLMs are trained on vast amounts of publicly available text, their parameters may contain information that was unavailable to investors at the time of the prediction task, potentially

leading to an overestimation of their real-time predictive performance. To address this concern, [2] evaluated their models on post-training data and applied anonymisation procedures to remove potential identifying information from the text. Similarly, [4] constructed chronologically consistent LLMs trained only on information available up to each prediction period and found that their results remained largely unchanged, suggesting that look-ahead bias is unlikely to be driving their findings.

Chapter 3

Methodology

This chapter begins with a description of the news dataset and the preprocessing steps used to prepare the data, followed by the construction of the "raw news" embeddings. Following this, we outline the methodology used to construct and interpret the "pure news" embeddings. The methodology is primarily motivated by the framework of [4], but the limited cross-section of stocks in our dataset makes the original cross-sectional residualisation procedure infeasible. We therefore develop an adapted regression framework to remove the component of the "raw news" embeddings associated with observable firm characteristics. Because the change in the residualisation framework may affect the interpretation of the resulting residual embedding, we finally investigate the source of the variation explained by the regression. The results are then used to determine what the residual embedding represents in this setting and to motivate the procedure used to construct the "pure news" embeddings employed in the remainder of the thesis.

3.1 Data Overview

This section provides an overview of the dataset used throughout this study. The dataset was obtained using *NewsWitch*, a product developed by ZanistaAI. Each observation in the dataset represents a news article associated with a stock and includes the corresponding date and timestamp information.

3.1.1 Dataset Fields

The dataset represents each news article using multiple textual fields and includes additional AI-generated analytical scores related to the article's characteristics. Table 3.1 provides an overview of the fields.

3.1.2 Data Coverage Period

The dataset consists of news articles associated with a universe of 14 stocks. The selected stocks are all large-cap companies drawn from the S&P 500, ranging from mega-cap firms exceeding $3 trillion in market capitalisation to firms closer to $450 billion, and span a range of industries and sectors. A complete list of the selected stocks, together with their respective industries and market capitalisations, is provided in Appendix B.1.

The initial dataset contains 594,627 news articles across the selected stocks. To ensure consistency in data coverage, the analysis is restricted to articles dated from January 2024 onwards. Earlier observations are excluded due to sparse coverage and limited representation across the selected stocks. The resulting dataset covers the period from January 2024 to July 2025.

Table 3.1: Variables in the news dataset.

Variable	Type	Description
`title`	Text	Headline of the article as returned by the discovery source.
`content`	Text	Full body text of the article.
`summary`	Text	Concise summary of the article focused on the tracked company or asset.
`direction`	Categorical	AI-estimated direction of the potential impact of the article on the tracked company or asset (positive / negative / neutral).
`sentiment_impact`	Numeric	AI-estimated magnitude and direction of the potential impact, represented on a scale from -10 to $+10$.
`intensity`	Categorical	AI-estimated time horizon of the potential effect of the article: *immediate*, *short-term*, *medium-term*, or *long-term*.
`relevance_score`	Numeric	AI-estimated measure of how directly the article relates to the tracked company or asset.
`reputation_score`	Numeric	AI-estimated score representing the reputation of the article source in relation to the news item.
`news_tags`	Categorical	AI-generated topic labels consisting of a primary theme and a supporting theme that describe the main subject of the article. A value of *None* indicates that no suitable topic tag was identified. The complete list of available tags is provided in the Appendix B.2.
`news_source_type`	Categorical	Category describing the type of publisher (e.g., financial media, company investor relations, regulators, research providers, or social media). This field categorises the publisher rather than the website domain. The complete list of source types is provided in the Appendix B.3.

3.1.3 Primary Text Field

The primary textual field used throughout this study is the `content` field, as it provides the highest coverage across the dataset. Therefore, unless stated otherwise, references to a news article throughout this study correspond to the text contained within this field.

3.1.4 Terminology

Throughout this study, several different levels of granularity are used to describe the news data and the subsequent analysis, and we introduce the corresponding terminology here. The most granular unit in the dataset is an individual *news article*, or simply *article*, which is associated with a specific stock, date, and timestamp. Since multiple articles may be published for the same stock at the same timestamp, we refer to the collection of articles sharing a common stock, date, and timestamp as a *stock-date-timestamp* observation. Similarly, articles associated with the same stock on the same calendar day are referred to collectively as a *stock-date* observation, while articles associated with the same stock within the same calendar month are referred to as a *stock-month* observation.

3.2 News Embedding Construction

This section describes the steps undertaken to construct the "raw news" embeddings. However, given the large size of the initial dataset, we first identify and remove duplicate and near-duplicate articles. The remaining articles are then converted into vector embeddings using the selected embedding model.

3.2.1 De-duplication

To avoid overweighting the same information, duplicate and near-duplicate news articles were removed following the procedure of [2]. Each article was first represented using a Bag-of-Words (BoW) vector representation, in which a document is represented by the occurrence of words while disregarding their order. Term Frequency-Inverse Document Frequency (TF-IDF) weighting was then applied to assign greater importance to informative terms and reduce the influence of commonly occurring words [14, Chapter 6, pp. 117, 119].

To quantify the similarity between the news articles, we used cosine similarity. Cosine similarity measures the similarity between two document vectors by computing the cosine of the angle between them, making it insensitive to document length [14, Chapter 6, p. 121]. Formally, the cosine similarity between two document vectors d_1 and d_2 is defined as

$$\text{Cosine Similarity}(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\|\|d_2\|}$$

where the numerator represents the dot product of the vectors d_1 and d_2, while the denominator is the product of their Euclidean norms[1].

For each article, only articles covering the same stock within the previous five days were considered as duplication candidates. We then computed cosine similarity between the current article and all candidates in this rolling window. Any article with a maximum cosine similarity ≥ 0.8 against a prior article was classified as a near-duplicate and removed. This procedure reduced the dataset by 42.3%, resulting in a dataset containing 343,085 articles.

[1]For a vector $d \in \mathbb{R}^n$, the Euclidean norm is defined as $\|d\|_2 = \left(\sum_{j=1}^{n} d_j^2\right)^{1/2}$.

3.2.2 Text Embedding Generation

News articles were converted into vector embeddings using OpenAI's text-embedding-3-large model, accessed through the Azure OpenAI API. The text-embedding-3-large model produces embeddings of dimension 3,072 for each article [19], which are used throughout the subsequent analysis. We refer to these embeddings as "raw news" embeddings.

3.3 News Embedding Residualisation

This section describes the methodology used to separate the component of the "raw news" embeddings associated with observable firm characteristics from the remaining component. We first prepare the embeddings and firm characteristics for the residualisation framework before introducing the characteristic-based regression motivated by [4]. We then adapt the regression procedure to our setting and evaluate its ability to explain variation in the embeddings. The section concludes by examining whether the variation identified by the characteristic-based approach primarily reflects observable firm characteristics or persistent differences across firms. This motivates the use of firm fixed effects and, thus, leads to a revised interpretation of the "pure news" embeddings.

3.3.1 Z-score Normalisation

Vector embeddings are not uniformly distributed in all directions. Rather, they take on a narrow cone shape in the vector space. This problem is called anisotropy [26]. According to [4], anisotropy can affect the effectiveness of embeddings for downstream analysis. There are several efficient methods in the existing literature to tackle this problem. For example, [26] applied a whitening transformation, which centres the embedding distribution by removing the mean and transforms the covariance matrix towards an identity matrix, resulting in a more isotropic representation. Another problem similar to anisotropy is the hubness problem seen in few-shot learning, where certain vectors become disproportionately close to many other vectors and frequently appear as nearest neighbours. [5] showed that Z-score normalisation can mitigate this issue by standardising the distribution of embedding dimensions. Following this approach, [4] applied a Z-score normalisation to all stock-month embeddings. For each month t, every embedding dimension was normalised by subtracting the pooled mean computed from all stock-month observations up to month $t-1$, and dividing by the corresponding pooled expanding standard deviation. However, our dataset differs from that considered in [4], as embeddings are not aggregated to a stock-month level. Instead, embeddings are defined for each individual article allowing multiple embeddings to be associated with the same stock within a month. However, for the downstream analysis, we would require embeddings at a stock-date level. Therefore, we adapt the normalisation procedure by applying an expanding-window Z-score transformation at the date level. For each date t, each embedding dimension is standardised using the pooled mean and standard deviation calculated from all observations available before date t.

From this point onwards, the z-score normalised "raw news" embeddings are used throughout this section. However, we refrain from referring to these embeddings as "raw news" embeddings to avoid confusion with the terminology used in Chapter 4, where the predictive performance of the "raw news" and "pure news" embeddings is compared. In that chapter, "raw news" refers specifically to the original, unnormalised embeddings. Therefore, throughout the remainder of this section, the z-score normalised "raw news" embeddings are referred to simply as "news embeddings" or "embeddings".

3.3.2 Firm Characteristics

The next step in the methodology is to remove the component of news embeddings that can be explained by observable firm characteristics. To achieve this, we use the characteristics dataset introduced by [7]. The JKP dataset consists of 153 firm characteristics constructed from equity data across 93 countries. These characteristics primarily capture a wide range of firm attributes. The complete list of characteristics can be found in the appendix of [7].

The use of JKP characteristics to analyse the information contained in news embeddings follows recent literature. [2] regressed stock returns predicted using LLM-based news embeddings on the 13 thematic clusters of JKP characteristics and showed that these characteristics explain a non-trivial proportion of the variation in embedding-based return predictions. Similarly, [4] regressed news embeddings directly on JKP characteristics and interpreted the residual component as the portion of news information that is orthogonal to observable firm characteristics, which they referred to as "pure news". Their procedure achieved a pooled R^2 (see Section 3.3.4 for the definition) of 7.7% when explaining embedding variation using JKP characteristics. The authors then showed that portfolios constructed using the residual embedding component generated significant return predictability.

Following this approach, we use JKP characteristics to purge the predictable component of each news embedding. The publicly available JKP characteristic data is obtained from the JKP Factors website [8]. The JKP characteristics for a stock are measured at a monthly frequency. Since the news dataset spans from January 2024 to July 2025, JKP characteristics from December 2023 to June 2025 are used. The one-month lag ensures that only information available prior to the news observation is used, thereby avoiding potential look-ahead bias. Since multiple articles may occur for the same stock within a given month, all embeddings corresponding to a stock-month period are assigned the same monthly JKP characteristic values.

Although the original JKP dataset contains 153 characteristics, not all characteristics have sufficient coverage for the stock universe considered in this study. Therefore, following the filtering procedure of [4], characteristics are ranked according to their availability and the 119 characteristics with the highest coverage are retained. The selected characteristics are then used as explanatory variables in the subsequent regression framework. A complete list of the retained characteristics and their economic interpretations is provided in Appendix B.4.

Moreover, the JKP characteristics are measured on different scales and therefore require standardisation before they can be used in the regression framework. Following [4], each characteristic is rank standardised cross-sectionally, i.e., for each month, stocks are ranked according to the value of each characteristic, and the resulting ranks are mapped to the interval $[-0.5, 0.5]$ according to

$$\tilde{x}_{i,t} = \frac{\text{rank}(x_{i,t}) - 1}{N_t - 1} - \frac{1}{2},$$

where $x_{i,t}$ denotes the value of a characteristic for stock i in month t, $\text{rank}(x_{i,t})$ is its cross-sectional rank, and N_t is the number of stocks in the cross-section for month t.

Rank standardisation is widely adopted in the asset pricing literature because the relative position of a firm within the cross-section is often more informative than the raw value of a characteristic. For example, the market capitalisation of a large firm has increased substantially over time, such that a firm valued at USD 1 billion would have been considered large several decades ago but relatively small today. Ranking firms within each month removes these time-varying scale effects and provides a consistent measure of a firm's relative characteristics across time [6]. In the context of this study, rank standardisation is also conceptually appropriate because the objective is to remove the

component of the news embeddings that is explained by a firm's relative characteristics, rather than their absolute numerical values.

Unlike [4], our stock universe consists of only 14 stocks. Performing the rank standardisation using such a small cross-section would produce only a limited number of distinct rank values, resulting in a coarse representation of the underlying characteristics. To obtain a finer and more stable ranking, the rank standardisation is instead performed using the entire cross-section of common stocks traded on the main U.S. exchanges. The standardised characteristic values corresponding to the 14 stocks in our sample are then retained for the subsequent analysis.

3.3.3 Embedding Residualisation Framework

This section introduces the regression framework used to separate the component of the news embeddings associated with observable firm characteristics from the remaining component. We first describe the cross-sectional specification proposed by [4], which provides the basis for the residualisation approach. We then adapt this framework to the pooled structure of our dataset. Given the high dimensionality of the characteristic space, we then specify a ridge-regularised framework to obtain a more stable estimate of the relationship between firm characteristics and news embeddings.

Original Cross-Sectional Characteristic Regression

Let $S_t \in \mathbb{R}^{N_t \times L}$ denote the matrix of firm characteristics at month t, where N_t is the number of stocks in the cross-section for month t, and L is the number of characteristics, including the intercept. Each row of S_t corresponds to a stock, while each column corresponds to a characteristic. Similarly, let $E_t \in \mathbb{R}^{N_t \times D}$ denote the corresponding matrix of news embeddings, where each row contains the embedding vector associated with a stock at month t, and D is the embedding dimension. We estimate the relationship between the firm characteristics and the embeddings using the cross-sectional regression

$$E_t = S_t \beta_t + \epsilon_t,$$

where $\beta_t \in \mathbb{R}^{L \times D}$ is the matrix of regression coefficients and $\epsilon_t \in \mathbb{R}^{N_t \times D}$ is the matrix of residuals. The ordinary least squares estimator is given by

$$\hat{\beta}_t = \left(S_t^\top S_t\right)^{-1} S_t^\top E_t,$$

provided that $S_t^\top S_t$ is invertible. The fitted component is therefore

$$\hat{E}_t = S_t \hat{\beta}_t,$$

which captures the portion of the embedding space explained by the firm characteristics. Now, the residual component,

$$\epsilon_t = E_t - \hat{E}_t,$$

represents the part of the embedding that cannot be explained by these characteristics.

Adapted Pooled Regression

Directly applying the regression in Section 3.3.3 to our dataset is not feasible. The original approach estimates a separate coefficient matrix for each month using the available cross-section of stocks. In our dataset, the number of observations ($N_t = 14$) available within an individual period is smaller than the number of firm characteristics ($L = 120$) included

in the regression, resulting in an underdetermined regression problem. Therefore, instead of estimating a separate coefficient matrix β_t for each month, we estimate a single β using the entire sample.

Let N denote the total number of news embedding observations in the dataset, where each observation corresponds to a unique article. Let L denote the number of JKP firm characteristics included in the regression, including the intercept, and let D denote the dimensionality of the news embeddings. The corresponding characteristic matrix is denoted by $S \in \mathbb{R}^{N \times L}$, where each row contains the JKP characteristics assigned to the corresponding news observation based on the stock and month of the embedding. As described previously, JKP characteristics are measured at a monthly frequency. Therefore, multiple news observations associated with the same stock within a given month share the same characteristic vector. The corresponding embedding matrix is denoted by $E \in \mathbb{R}^{N \times D}$, where each row contains the news embedding associated with a particular article.

The pooled regression model is therefore given by

$$E = S\beta + \epsilon, \tag{3.3.1}$$

where $\beta \in \mathbb{R}^{L \times D}$ is a coefficient matrix capturing the relationship between firm characteristics and embedding dimensions across the full sample, and $\epsilon \in \mathbb{R}^{N \times D}$ represents the residual embedding component. Further, the ordinary least squares (OLS) estimate, $\hat{\beta}_{OLS}$ is given by:

$$\hat{\beta}_{OLS} = (S^\top S)^{-1} S^\top E \tag{3.3.2}$$

provided that $S^\top S$ is invertible.

Ridge Regression

Due to the high dimensionality of the characteristic space and the presence of correlated predictors, ordinary least squares estimation in Section 3.3.3 may lead to unstable coefficient estimates. This issue is particularly relevant for the JKP characteristics, where many characteristics capture related aspects of firm behaviour and may therefore exhibit substantial multicollinearity. To address this issue, we estimate the characteristic-to-embedding mapping using ridge regression, which introduces an L_2 regularisation penalty on the regression coefficients.

The ridge estimator is obtained by solving:

$$\hat{\beta}_{ridge} = \arg\min_{\beta} \left\{ \|E - S\beta\|_F^2 + \lambda \operatorname{tr}(\beta^\top P \beta) \right\}$$

where λ controls the strength of the regularisation penalty, $\|\cdot\|_F$ denotes the Frobenius norm[2], $\operatorname{tr}(\cdot)$ denotes the trace operator, and P is the penalty matrix controlling which coefficients are regularised. Since we choose not to penalise the intercept, P is defined as

$$P = \begin{bmatrix} 0 & 0 \\ 0 & I_{L-1} \end{bmatrix},$$

where the first row and column correspond to the intercept term and I_{L-1} is the identity matrix for the remaining $L-1$ characteristics. This formulation applies the L_2 penalty only to the characteristic coefficients while leaving the intercept unregularised.

The corresponding closed-form ridge estimator is

$$\hat{\beta}_{ridge} = (S^\top S + \lambda P)^{-1} S^\top E. \tag{3.3.3}$$

[2]For a matrix $A \in \mathbb{R}^{m \times n}$, the Frobenius norm is defined as $\|A\|_F = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij}^2 \right)^{1/2}$.

The regularisation term reduces the sensitivity of the coefficient estimates to multicollinearity and improves the stability of the estimated characteristic-to-embedding mapping.

Using the estimated coefficients, the component of the embeddings explained by firm characteristics is calculated as:

$$\hat{E} = S\hat{\beta}_{ridge},$$

and the residual component is obtained as:

$$E^{pure} = E - \hat{E}.$$

The resulting matrix $E^{pure} \in \mathbb{R}^{N \times D}$ represents the component of the news embeddings that is not explained by the observable firm characteristics under the regularised linear projection. Following [4], we initially refer to this residual representation as the "pure news" embedding. But we interpret the residual embedding as the embedding variation that remains unexplained after accounting for the regularised relationship between firm characteristics and news embeddings. In the coming sections, a more economic interpretation is provided.

3.3.4 Empirical Evaluation

In this section, we evaluate the adapted pooled regression framework described in Section 3.3.3. The regression performance is not evaluated with the objective of maximising explanatory power, but rather with the objective of quantifying the extent to which variation in news embeddings can be attributed to observable firm characteristics, as measured by the coefficient of determination (R^2). Since the residual embeddings are obtained by removing the component explained by these regressions, this section provides an assessment of how much of the original embedding representation would be removed under the different regression specifications before constructing the residual embeddings.

We first estimate the ordinary least squares baseline using the estimator defined in Equation 3.3.2. The results from this baseline specification motivate the use of ridge regression, as introduced in Section 3.3.3. We then examine the selection of the regularisation parameter λ in the ridge estimator defined in Equation 3.3.3 and discuss the results obtained from the regularised regression framework.

Ordinary Least Squares Baseline

First, we estimate the adapted pooled regression model introduced in Equation 3.3.1 using the ordinary least squares estimator defined in Equation 3.3.2. This regression provides a baseline to quantify the variation in the news embeddings explained by the firm characteristics. In addition to the regression using the JKP characteristics, we consider a placebo benchmark in which each characteristic is replaced with values sampled independently from a uniform distribution on the interval $[-0.5, 0.5]$. The purpose of this benchmark is to assess whether the explanatory power of the regression reflects a genuine relationship between the firm characteristics and the news embeddings, rather than arising from overfitting in a high-dimensional regression setting.

To evaluate the explanatory power of each specification, we report an adjusted pooled R^2, which aggregates the residual and total sums of squares across all embedding coordinates before applying the degrees-of-freedom adjustment. Let D denote the embedding dimension, $E_j^{(c)}$ the cth coordinate of the embedding vector for observation j, and $\widehat{E}_j^{(c)}$ its fitted value from the OLS regression. The adjusted pooled R^2 is defined as

$$R^2_{\text{pooled}} = 1 - \frac{\sum_{c=1}^{D}\sum_{j=1}^{N}\left(E_j^{(c)} - \widehat{E}_j^{(c)}\right)^2/(N-L)}{\sum_{c=1}^{D}\sum_{j=1}^{N}\left(E_j^{(c)}\right)^2/N}, \tag{3.3.4}$$

where N is the total number of observations, L is the number of estimated regression parameters including the intercept, and $D = 3,072$ is the embedding dimension. Further, to examine how the explanatory power varies across individual embedding coordinates, we compute a coordinate-wise adjusted R^2. For embedding coordinate c, this is defined as

$$R^2(c) = 1 - \frac{\sum_{j}^{N}\left(E_j^{(c)} - \widehat{E}_j^{(c)}\right)^2/(N-L)}{\sum_{j}^{N}\left(E_j^{(c)}\right)^2/N}, \qquad c = 1, \ldots, D.$$

Using the JKP characteristics, the pooled regression achieves an adjusted pooled R^2 of 15.41%. In comparison, the placebo regression using random characteristics achieves an adjusted pooled R^2 of 9.34%. Figure 3.1 presents the distributions of these coordinate-wise adjusted R^2 values across the embedding dimensions for both the JKP and placebo regressions.

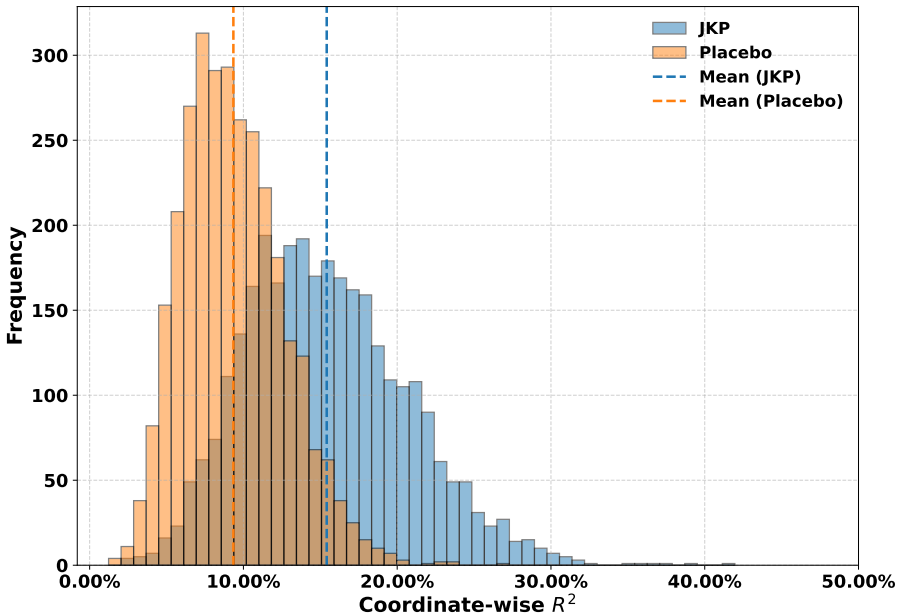


Figure 3.1: Distribution of coordinate-wise adjusted R^2 values across the 3,072 embedding dimensions using JKP characteristics and the placebo benchmark.

The pooled regression explains approximately 15.41% of the variation in the news embeddings, which is higher than the approximately 7% reported by [4]. However, this comparison should be interpreted with caution, as the two datasets differ substantially in both their construction and dimensionality. In particular, our dataset contains only 14 firms, while the regression includes 120 explanatory variables, increasing the risk of overfitting. This is reflected in the placebo benchmark, where random characteristics alone explain approximately 9.34% of the embedding variation. Nevertheless, the JKP

characteristics explain an additional 6% of the variation beyond the placebo benchmark, suggesting that the regression captures structure in the embeddings that is not attributable solely to the high-dimensional nature of the regression.

Figure 3.1 shows that the coordinate-wise explanatory power is distributed across many embedding dimensions in both the JKP and placebo regressions. The higher explanatory power of the JKP regression relative to the placebo benchmark is therefore not attributable to a small subset of coordinates alone. However, these in-sample results are insufficient to determine whether the regression captures a stable relationship between firm characteristics and news embeddings or is influenced by overfitting. To better understand the nature of the variation captured by the regression, we next investigate whether this relationship generalises across time.

Expanding-Window Validation of the OLS Regression

To evaluate whether the relationship estimated by the pooled ordinary least squares estimator in Equation 3.3.2 generalises over time, we perform an expanding-window evaluation. In this method, the training sample is recursively expanded over time. At each step, all observations up to month $t-1$ are used to estimate the model, and the estimated relationship is evaluated on month t. By restricting the estimation sample to information available prior to the evaluation month, this procedure avoids look-ahead bias. This is particularly important because the residual embeddings should be constructed using only information that would have been available at the time of each observation, rather than benefiting from relationships estimated using future data. Furthermore, the earliest evaluation months are omitted since the expanding training sample is initially too small to estimate the pooled regression reliably. To be precise, the number of training observations must exceed the number of estimated parameters and the design matrix must have full rank before estimation is performed. During the earliest months of the sample, these conditions are not satisfied due to the limited number of available observations.

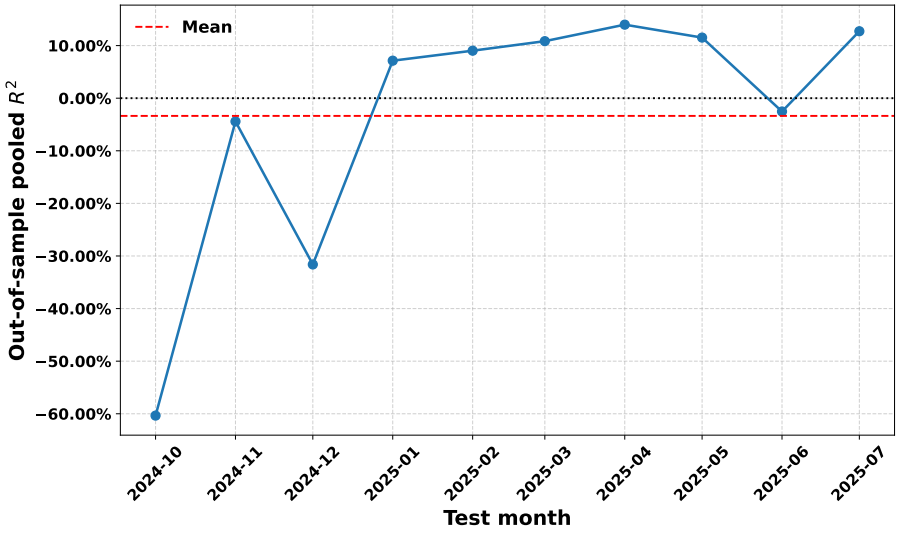
The expanding-window evaluation is carried out for both the regression using the JKP firm characteristics and the placebo benchmark constructed from random characteristics. The performance is evaluated using the pooled out-of-sample R^2. For each evaluation month t, the pooled out-of-sample R^2 is defined as

$$R^2_{\text{OOS},t} = 1 - \frac{\sum_{c=1}^{D} \sum_{j \in \mathcal{T}_t} \left(E_j^{(c)} - \widehat{E}_j^{(c)} \right)^2}{\sum_{c=1}^{D} \sum_{j \in \mathcal{T}_t} \left(E_j^{(c)} \right)^2},$$

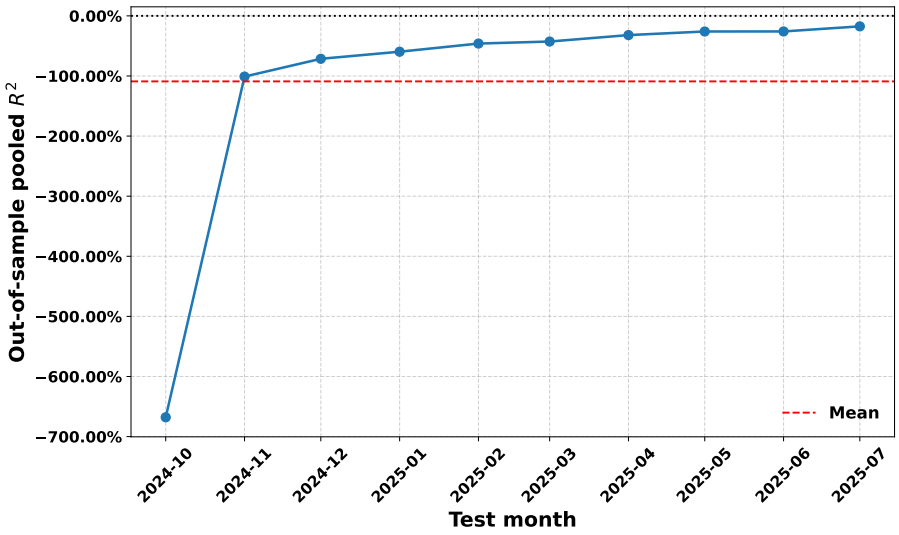
where $D = 3,072$ is the embedding dimension, $\mathcal{T}_t$ denotes the set of observations in evaluation month t, $E_j^{(c)}$ is the observed value of embedding coordinate c for observation j, and $\widehat{E}_j^{(c)}$ is the corresponding out-of-sample fitted value obtained using regression coefficients estimated exclusively from observations in the expanding training window preceding month t. Unlike Equation 3.3.4 used for in-sample evaluation, no degrees-of-freedom adjustment is applied here because the evaluation observations are held out from model estimation.

Figure 3.2a and Figure 3.2b illustrate the resulting monthly pooled out-of-sample R^2 values for the regression using JKP firm characteristics and the placebo benchmark, respectively.

The results indicate that the ordinary least squares regression exhibits considerable instability when evaluated out-of-sample. Although the regression using the JKP characteristics gradually achieves positive out-of-sample R^2 values as the training sample expands,



(a) JKP characteristics



(b) Placebo benchmark

Figure 3.2: Monthly pooled out-of-sample R^2 values from the expanding-window ordinary least squares regression.

performance is poor during the earliest evaluation periods and remains variable throughout the sample, with a mean pooled out-of-sample R^2 of -3.36%. On the other hand, the placebo benchmark produces consistently negative out-of-sample R^2 values across all evaluation months, suggesting that the additional explanatory power obtained from the JKP characteristics is not reproduced by a set of randomly generated predictors. Nevertheless, this instability of the OLS estimates motivates the introduction of ridge regression as a regularised alternative.

Selection of the Ridge Regularisation Parameter

To improve the stability of the pooled regression, we consider the ridge estimator defined in Equation 3.3.3. The regularisation parameter λ is selected using the same expanding-window framework described in Section 3.3.4, such that the choice is based entirely on out-of-sample performance. Unlike the OLS estimator, ridge regression does not require the characteristic matrix S to have full column rank. As shown in Equation 3.3.3, the regularisation term λP modifies the matrix being inverted from $S^\top S$ to $S^\top S + \lambda P$, improving numerical stability and allowing estimation in periods where the unregularised OLS formulation is affected by rank deficiencies.

The candidate values considered are

$$\lambda \in \{0, 1, 10, 100, 1000, 3000, 5000, 10000\}.$$

For each value of λ, the ridge regression is estimated within every expanding-window training sample and evaluated on the subsequent month. Similar to Section 3.3.4, the standard pooled out-of-sample R^2 is used as the selection criterion. The optimal regularisation parameter is chosen as the value that maximises the mean pooled out-of-sample R^2 across all evaluation months.

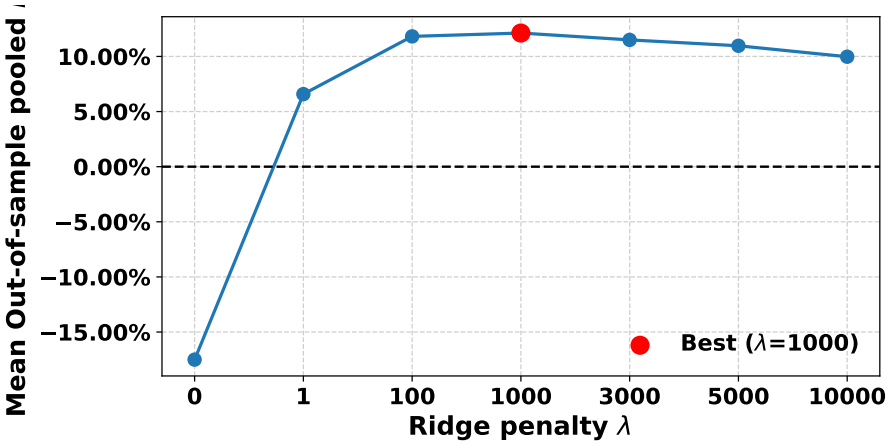
Figure 3.3a presents the mean pooled out-of-sample R^2 obtained for the regression using JKP characteristics across the candidate values of λ. The corresponding placebo results are reported in Figure 3.3b.

For the regression using the JKP characteristics, increasing the regularisation strength substantially improves out-of-sample performance relative to the ordinary least squares estimator. Performance increases steadily before reaching its maximum at $\lambda = 1000$, after which stronger regularisation gradually reduces explanatory power. Although regularisation also improves the performance of the placebo benchmark, its out-of-sample R^2 remains negative across all values of λ, indicating that regularisation alone does not generate meaningful explanatory power. Hence, $\lambda = 1000$ is selected as the regularisation parameter for the remainder of this study.

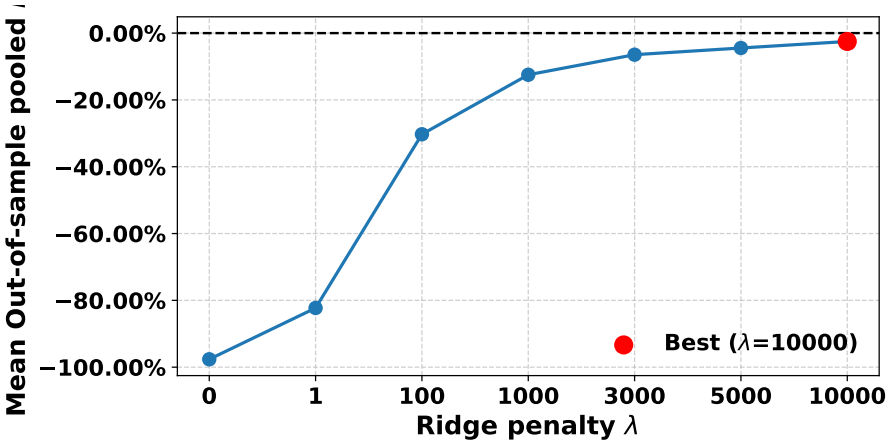
Expanding-Window Diagnostic of the Ridge Regression

In this section, the results of the expanding-window evaluation using ridge regression for the selected regularisation parameter $\lambda = 1000$ are presented.

Relative to the ordinary least squares estimator, ridge regression produces consistently positive out-of-sample R^2 values across all evaluation months, achieving a mean and a median pooled out-of-sample R^2 of 12.12% and 12.19%, respectively, across the evaluation period. The minimum pooled out-of-sample R^2 is 5.39%, occurring during the earliest evaluation period when the available training sample is smallest, while the maximum is 15.14%. Figure 3.4 presents the corresponding monthly pooled out-of-sample R^2 values. As one can see, the substantial variability observed under the OLS specification is largely reduced, resulting in more stable performance throughout the evaluation period. These findings indicate that the ridge penalty improves the stability of the estimated regression while preserving the explanatory structure captured by the observable firm characteristics.



(a) JKP characteristics



(b) Placebo benchmark

Figure 3.3: Mean pooled out-of-sample R^2 across candidate ridge regularisation parameters λ.

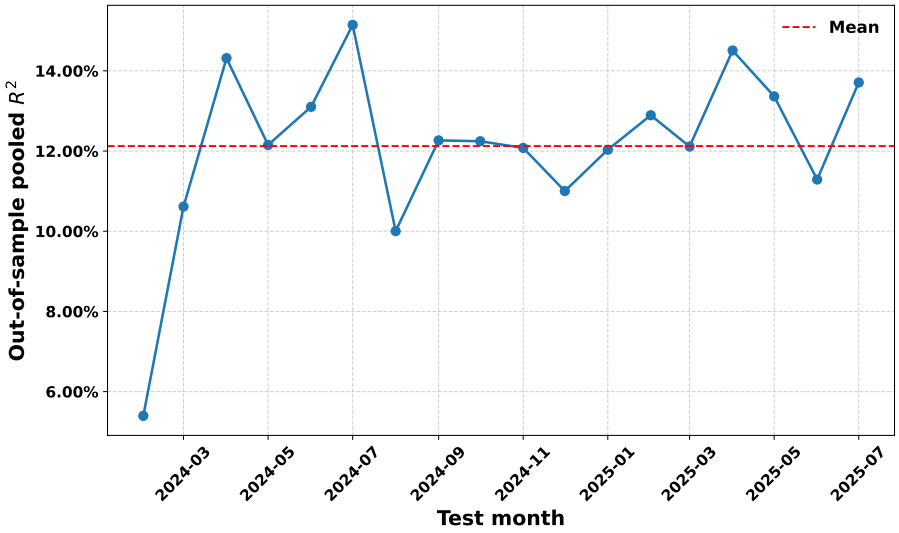


Figure 3.4: Monthly pooled out-of-sample R^2 values from the expanding-window ridge regression using $\lambda = 1000$.

Much care should be taken in interpreting these values. The out-of-sample R^2 values should be interpreted as a diagnostic assessment of the stability of the decomposition, quantifying the proportion of embedding variation that can be attributed to observable firm characteristics when the relationship is estimated using only past information. The expanding-window procedure ensures that there is no look-ahead bias during estimation or evaluation. However, these values should not be interpreted as unbiased estimates of predictive performance, as the regularisation parameter was selected using the expanding-window procedure primarily to obtain a stable decomposition rather than to maximise out-of-sample prediction accuracy.

3.3.5 Interpreting the Explained Variation

Section 3.3.4 established that the adapted pooled regression explains a meaningful proportion of the variation in the news embeddings and that this relationship generalises under regularised estimation. However, before constructing the residual embeddings using the selected regularisation parameter, it is important to understand not only how much variation is removed, but also what information this variation represents. The coefficient of determination (R^2) quantifies the amount of embedding variation explained by the regression, but it does not reveal the nature of that explained variation. The objective of this section is therefore to investigate the source of the variation explained by the regression and to provide an economic interpretation of the information removed when constructing the residual embeddings.

We begin by performing a robustness check in which the association between firms and their characteristic vectors is deliberately disrupted. The results of this experiment motivate a more detailed investigation into the source of the regression's explanatory power. We then introduce a firm fixed effects benchmark and perform a series of robustness checks to compare the characteristic-based regression with a specification that captures only persistent firm-level differences. Collectively, these analyses indicate that the variation explained by the regression is predominantly attributable to firm identity, providing a clearer interpretation of the residual embedding representation.

Permutation Test

To analyse the source of the variation explained by the regression, we perform a permutation test in which the correspondence between stock-months and their associated firm characteristics is deliberately disrupted. Since all news embeddings belonging to the same stock within a given month share the same JKP characteristic vector, the permutation is carried out at the stock-month level rather than at the individual article level, i.e., each stock-month retains its original news embeddings, while its entire vector of JKP characteristics is replaced with the characteristic vector from another randomly selected stock-month. Thus, all news embeddings belonging to a given stock-month continue to share the same characteristic vector, but this vector no longer corresponds to the firm's true characteristics.

Formally, let S^{π} denote the permuted characteristic matrix obtained by randomly reassigning the rows of the original characteristic matrix S in Equation 3.3.1. Specifically, for an individual stock-month observation, the characteristic vector assigned to firm i in month t is replaced by the characteristic vector from another randomly selected firm j in month τ, i.e.,

$$S^{\pi}_{i,t} = S_{j,\tau},$$

where $(j, \tau) \neq (i, t)$. The regression is then estimated using:

$$E = S^{\pi}\beta + \epsilon.$$

This preserves the empirical distribution of the characteristics and the dependence structure among characteristics while removing the original correspondence between firm characteristics and their associated news embeddings.

Following the permutation, we estimate the adapted pooled regression model introduced in Equation 3.3.1 using the ordinary least squares estimator defined in Equation 3.3.2. The adjusted pooled R^2 decreases from 15.41% to 9.44%.

The substantial reduction in explanatory power indicates that the regression relies on the original correspondence between firm characteristics and the associated news embeddings. A notable observation is that the resulting R^2 values are very similar to those obtained using random characteristics in Section 3.3.4. This indicates that once the correspondence between firms and their characteristic vectors is disrupted, the regression no longer captures meaningful explanatory variation beyond what would be obtained from random inputs. Therefore, the explanatory power observed in the original specification is attributable to the firm-specific assignment of characteristics rather than the marginal distribution of the characteristics or the high-dimensional regression setting. This suggests that a considerable proportion of the variation removed by the regression may be associated with persistent firm-level information.

Firm Fixed Effects Benchmark

In this section, we introduce a firm fixed effects benchmark. This specification captures the extent to which persistent differences between firms explain variation in the news embeddings, without incorporating any firm characteristics. In this benchmark, the predicted embedding for each observation is obtained from the average embedding of the corresponding firm within the estimation sample.

Formally, let $E_{i,j}$ denote the news embedding associated with the jth news observation of firm i. The firm fixed effects specification assumes that the embedding can be decomposed as:

$$E_{i,j} = \alpha_i + \epsilon_{i,j},$$

where α_i represents the persistent firm-specific component of the embedding and $\epsilon_{i,j}$ captures the remaining observation-level variation. The estimated firm effect is obtained as the average embedding of firm i over all observations available in the estimation sample:

$$\hat{\alpha}_i = \frac{1}{N_i} \sum_{j \in \mathcal{J}_i} E_{i,j},$$

where $\mathcal{J}_i$ denotes the set of news observations associated with firm i in the estimation sample, and $N_i = |\mathcal{J}_i|$ represents the number of observations in this set. The predicted embedding is therefore given by:

$$\hat{E}_{i,j} = \hat{\alpha}_i.$$

The explanatory power of this benchmark measures the proportion of embedding variation that can be attributed to persistent firm-level differences. The adjusted pooled R^2 of the firm fixed effects specification is defined as:

$$R^2_{\text{FE,adj}} = 1 - \frac{\sum_{c=1}^{D} \sum_{i} \sum_{j \in \mathcal{J}_i} \left(E_{i,j}^{(c)} - \widehat{\alpha}_i^{(c)} \right)^2 / (N - F)}{\sum_{c=1}^{D} \sum_{i} \sum_{j \in \mathcal{J}_i} \left(E_{i,j}^{(c)} \right)^2 / N},$$

where $D = 3,072$ is the embedding dimension, N is the total number of observations, F is the number of firms, $E_{i,j}^{(c)}$ denotes the cth coordinate of the embedding for observation j of firm i, and $\widehat{\alpha}_i^{(c)}$ denotes the cth coordinate of the estimated firm-specific effect for firm i.

The in-sample explanatory power of the firm fixed effects specification is comparable to that obtained from the characteristic-based regressions. The model achieves an adjusted pooled R^2 of 14.92%, compared with 15.41% for the OLS specification.

To compare the results with the ridge specification, we evaluate the firm fixed effects benchmark using the same expanding-window procedure described in Section 3.3.4. At each evaluation period, firm-specific average embeddings are estimated exclusively using observations contained in the training window. These estimated firm effects are then used to predict the embeddings of news observations in the subsequent evaluation month. This procedure ensures that the benchmark does not use future information when constructing firm-level embeddings, thereby avoiding look-ahead bias.

The results show that the firm fixed effects benchmark achieves substantially more stable out-of-sample performance compared with the ordinary least squares specification. Across the evaluation period, the model achieves a mean pooled out-of-sample R^2 of 15.02%, with values ranging from 12.51% to 17.44%. Figure 3.5 presents the monthly pooled out-of-sample R^2 values, demonstrating that the firm fixed effects benchmark maintains consistently positive performance throughout the evaluation period.

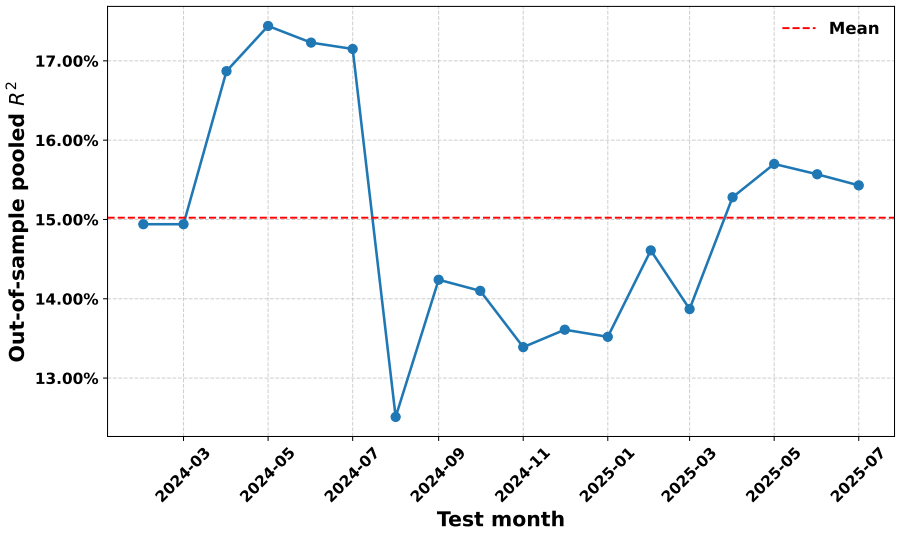


Figure 3.5: Monthly pooled out-of-sample R^2 values from the firm fixed effects benchmark evaluated using the expanding-window procedure.

These results show that persistent firm-level differences explain a larger and more stable proportion of the variation in the news embeddings compared to the characteristic-based regression specifications. Moreover, the higher out-of-sample performance relative to the characteristic-based regressions suggests that firm identity captures information beyond the observable JKP characteristics. However, similar explanatory power does not imply

that the two specifications capture identical components of the embedding representation. The variation explained by the firm fixed effects benchmark and the characteristic-based regressions may still differ, which motivates further analysis of the source of the explained variation.

Within-Firm Ridge Regression

In this section, we investigate whether the relationship between observable firm characteristics and the news embeddings remains after removing persistent firm-level differences. To do so, the ridge regression is re-estimated using firm-demeaned embeddings as the dependent variables. This decomposition isolates the within-firm component of the embeddings by removing the average embedding associated with each firm.

Formally, let $E_{i,j}$ denote the embedding associated with the jth news observation of firm i, and let $\mathcal{J}_i$ denote the set of observations available for firm i in the estimation sample. The average embedding of firm i is given by

$$\bar{E}_i = \frac{1}{N_i} \sum_{j \in \mathcal{J}_i} E_{i,j},$$

where $N_i = |\mathcal{J}_i|$ denotes the number of observations belonging to firm i. The firm-demeaned embedding is then defined as

$$\tilde{E}_{i,j} = E_{i,j} - \bar{E}_i.$$

The transformed embeddings therefore contain only within-firm variation, with the persistent firm-specific component removed. The ridge regression introduced in Section 3.3.3 is subsequently re-estimated using the demeaned embeddings,

$$\tilde{E} = S\beta + \epsilon,$$

where $\tilde{E}$ denotes the matrix whose rows are the demeaned embeddings.

To avoid look-ahead bias, the firm demeaning procedure is implemented using an expanding-window evaluation framework, i.e., at each evaluation month, the firm-specific average embeddings are estimated using only observations contained in the expanding training sample. These training-based averages are then used to demean both the training embeddings and the corresponding observations in the evaluation month, thereby ensuring that no future information is used during either the demeaning procedure or the regression estimation. Ridge regression is then estimated on the demeaned training embeddings and evaluated on the demeaned embeddings in the subsequent test month using the regularisation parameter $\lambda = 1000$.

The expanding-window results, presented in Figure 3.6, indicate that, after removing the persistent firm-level component of the embeddings, the characteristic-based regression has virtually no remaining explanatory power. Across the evaluation period, the demeaned ridge regression achieves a mean pooled out-of-sample R^2 of -0.27%, with values fluctuating around zero throughout the sample. This contrasts sharply with the positive out-of-sample performance obtained in Figure 3.4.

These findings suggest that the relationship between the JKP characteristics and the news embeddings is driven almost entirely by persistent firm-level differences. Once this firm-specific component is removed, the observable characteristics provide essentially no additional explanatory power for the remaining within-firm embedding variation. Combined with the strong performance of the firm fixed effects benchmark in the previous section, this indicates that the variation explained by the characteristic-based regression largely reflects persistent differences across firms rather than systematic within-firm changes associated with observable characteristics.

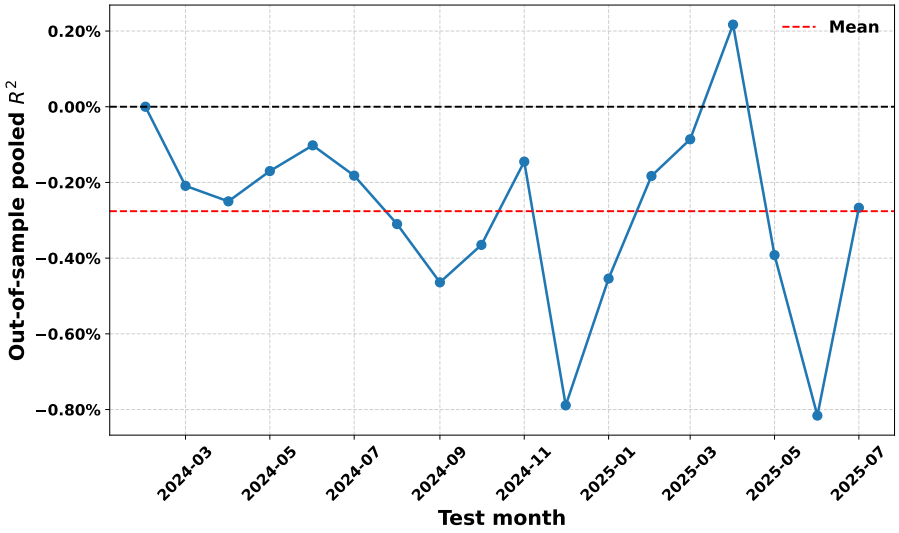


Figure 3.6: Monthly pooled out-of-sample R^2 values from the within-firm ridge regression evaluated using the expanding-window procedure with $\lambda = 1000$.

3.3.6 "Pure News" Embedding Construction

Based on the analyses presented in Sections 3.3.4 and 3.3.5, the residual news embeddings used throughout the remainder of this study are constructed by removing the persistent firm-level component from the original news embeddings. The ridge regression framework introduced in Section 3.3.3 was initially used to remove the component of the embedding variation explained by observable JKP firm characteristics. The regression coefficients were estimated within an expanding-window framework using the regularisation parameter $\lambda = 1000$, selected based on the expanding-window validation procedure described in Section 3.3.4.

However, the results presented in Section 3.3.5 show that the explanatory power of the characteristic-based regression is driven primarily by persistent firm-level differences. As a result, the firm fixed effects benchmark achieves comparable explanatory power to the characteristic-based regression, while the regression using firm-demeaned embeddings demonstrates that observable characteristics provide essentially no additional explanatory power once persistent firm-level variation has been removed. Therefore, the component removed by the characteristic regression is interpreted primarily as firm-specific information rather than time-varying information related to observable characteristics.

Hence, we use the firm fixed effects to construct the residual news embeddings. The residualisation procedure follows the expanding-window method to avoid look-ahead bias. At each evaluation period, the firm-specific embedding component is estimated using only observations available in the expanding training sample. The estimated firm effect is given by

$$\hat{\alpha}_i = \frac{1}{N_i} \sum_{j \in \mathcal{J}_i} E_{i,j},$$

where $\mathcal{J}_i$ denotes the set of observations associated with firm i in the training sample. The residual news embedding is then constructed as

$$E_{i,j}^{\text{pure}} = E_{i,j} - \hat{\alpha}_i.$$

This procedure removes the persistent firm-specific component while preserving the remaining observation-level variation in the news representation.

Thus, although the ridge residualisation framework is used to study the relationship between firm characteristics and embedding variation, the firm fixed effects residual embeddings are used as the primary representation of "pure news" throughout the remainder of this study. Accordingly, we interpret the resulting residual embeddings as a representation of news content that is largely independent of persistent firm identity, rather than simply the component orthogonal to observable firm characteristics.

An important point to note is that, as mentioned in Section 3.1.2, the dataset spans the period from January 2024 to July 2025. However, due to the requirements of the expanding-window method, residual embeddings cannot be computed for the initial month of the sample. Thus, all subsequent analyses cover the period from February 2024 to July 2025.

Chapter 4

Predicting Absolute Abnormal Returns

Chapter 3 introduced the construction of the "pure news" embeddings and discussed their interpretation. Having established these embeddings, the natural next question is whether they contain predictive information about future market movements. This chapter addresses this question by evaluating the forecasting performance of both the "pure news" and "raw news" embeddings in predicting future absolute abnormal returns.

The motivation for using absolute abnormal returns as the prediction target is to capture whether a news article is associated with a subsequent movement in the stock price, irrespective of the direction of that movement. Viewing the publication of a news article as an event that may be followed by a firm-specific price response, the objective is to determine whether the information contained in the article can predict whether the stock experiences a larger or smaller movement than would otherwise be expected given broader market conditions. We therefore abstract from the direction of the return and focus only on its magnitude. This provides a relatively simple test of whether the news embeddings contain information about subsequent stock-price movements, without requiring the model to predict whether the price will increase or decrease.

However, the initial and more straightforward objective was to predict signed returns. But this specification did not produce meaningful positive out-of-sample R^2 values in our analysis. At first sight, this may appear to contrast with the findings of [2], who report positive out-of-sample predictability for one-day signed returns. However, the two results are not directly comparable. [2] primarily evaluates return predictions using cross-sectional correlations between predicted and realised returns, rather than assessing the proportion of variation in individual returns explained by the predictions through out-of-sample R^2. This choice is particularly well suited to their subsequent objective of constructing long-short portfolios, for which correctly ranking stocks according to their expected returns is directly relevant. A positive cross-sectional correlation therefore indicates useful information for portfolio formation, but does not by itself indicate how much of the variation in individual stock returns is explained by the predictions. Hence, the absence of positive out-of-sample R^2 in our analysis does not necessarily contradict the positive predictive correlations reported in [2]. A further important difference is the substantially larger dataset considered in [2], both in terms of the time period and the cross-section of stocks, which provides considerably more observations over which relatively weak cross-sectional signals can be aggregated into portfolio-level performance.

The chapter begins by defining the prediction target, followed by the regression framework and the experimental design employed throughout the analysis. The predictive performance of both "raw news" and "pure news" embeddings is then evaluated and compared. This is followed by an examination of whether the observed results are consistent

across different news source types and news categories. Finally, we discuss the factors that may explain the differences in predictive performance between the embeddings.

4.1 Absolute Abnormal Returns

Stock returns reflect a combination of firm-specific information and aggregate market movements. Therefore, a movement in a stock price does not necessarily represent a firm-specific reaction, as the observed return may simply reflect broader market conditions affecting all stocks simultaneously. To remove the component of returns explained by aggregate market movements, a market model is estimated. The remaining residual component is referred to as the abnormal return [13].

To formally define this metric, it is first necessary to formally introduce returns. Let $P_{i,t}$ denote the adjusted closing price of stock i at trading day t. The simple return over a horizon of h trading days is defined as:

$$R_{i,t}^{(h)} = \frac{P_{i,t+h} - P_{i,t}}{P_{i,t}},$$

where h represents the forecasting horizon. In this study, however, returns are measured using logarithmic returns, which are widely used in empirical asset pricing applications. The logarithmic return of stock i over a horizon of h trading days is defined as:

$$r_{i,t}^{(h)} = \log(P_{i,t+h}) - \log(P_{i,t}).$$

To remove the component of stock returns associated with aggregate market movements, the corresponding market return is introduced. Let $P_{m,t}$ denote the value of any market index at day t. The logarithmic market return over the same horizon is defined as:

$$r_{m,t}^{(h)} = \log(P_{m,t+h}) - \log(P_{m,t}).$$

The relationship between stock returns and market movements is then estimated using a single-factor market model:

$$r_{i,t}^{(h)} = \alpha_i^{(h)} + \beta_i^{(h)} r_{m,t}^{(h)} + \epsilon_{i,t}^{(h)}, \tag{4.1.1}$$

where $\alpha_i^{(h)}$ represents the stock-specific intercept, $\beta_i^{(h)}$ captures the sensitivity of stock i to market movements, and $\epsilon_{i,t}^{(h)}$ represents the component of returns unexplained by market movements [13].

The estimated residual from this regression is defined as the abnormal return [13]:

$$AR_{i,t}^{(h)} = \hat{\epsilon}_{i,t}^{(h)} = r_{i,t}^{(h)} - \left(\hat{\alpha}_i^{(h)} + \hat{\beta}_i^{(h)} r_{m,t}^{(h)}\right). \tag{4.1.2}$$

The prediction target in this study is not the signed abnormal return but its absolute value:

$$\left|AR_{i,t}^{(h)}\right| = \left|\hat{\epsilon}_{i,t}^{(h)}\right|.$$

The absolute abnormal return, $|AR_{i,t}^{(h)}|$, therefore captures the magnitude of the stock-specific movement experienced by stock i at day t over the horizon h days, after accounting for aggregate market movements. A higher value indicates that the stock experienced a larger movement relative to what would be expected given overall market conditions, whereas a lower value indicates that the observed movement was closer to the level implied by market movements.

4.2 Regression Framework

The next step is to estimate the relationship between absolute abnormal returns and the embedding vectors. To evaluate this relationship, an ordinary least squares (OLS) regression framework is employed. This section formally describes the regression model and the dimensionality reduction procedure applied before estimation. Since the objective is to compare the predictive relationship between the "raw news" and "pure news" embeddings, both embedding representations are passed through the same regression framework described in this section. For simplicity, we use the term "embedding vector" to refer to either the "raw news" or "pure news" embedding vector.

Let $x_{i,t} \in \mathbb{R}^d$ denote the embedding vector associated with stock i on day t, where d represents the original embedding dimensionality. Due to the high dimensionality of the embedding space, directly estimating a regression model using the original dimensions may lead to overfitting and unstable parameter estimates. Therefore, principal component analysis (PCA) is applied to transform the embedding vectors into a lower-dimensional space [10]. For this, the embedding vectors are first standardised by removing the mean and scaling each dimension by its standard deviation. The mean embedding vector is defined as:

$$\bar{x} = \frac{1}{N} \sum_{i,t} x_{i,t},$$

where N denotes the number of stock-date observations. The standardised embedding vector is then defined as:

$$\tilde{x}_{i,t}^{(l)} = \frac{x_{i,t}^{(l)} - \bar{x}^{(l)}}{s^{(l)}},$$

where l indexes the embedding dimensions, $\bar{x}^{(l)}$ represents the mean of dimension l, and $s^{(l)}$ denotes the corresponding standard deviation.

Using the standardised embedding vectors, PCA identifies the directions of maximum variance by decomposing the covariance matrix:

$$\Sigma = \frac{1}{N-1} \sum_{i,t} \tilde{x}_{i,t} \tilde{x}_{i,t}^{\top}.$$

The principal component directions are obtained from the eigenvalue decomposition:

$$\Sigma w_j = \lambda_j w_j,$$

where w_j denotes the eigenvector associated with eigenvalue λ_j. The eigenvectors are ordered such that:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d,$$

with the first components capturing the largest amount of variation in the embedding space. Let:

$$W_k = [w_1, w_2, \ldots, w_k]$$

denote the matrix containing the first k principal component directions. The projection of each embedding vector onto this reduced space is then given by:

$$z_{i,t} = W_k^{\top} \tilde{x_{i,t}}, \tag{4.2.1}$$

where $z_{i,t} \in \mathbb{R}^k$ is the reduced-dimensional representation of the original embedding vector $x_{i,t}$.

Following the dimensionality reduction, an ordinary least squares (OLS) regression model is estimated using the k dimensions of $z_{i,t}$ as explanatory variables. The stock-date observations are stacked into a single training sample. Let j index the observations in this stacked dataset, where each observation j corresponds to a unique stock-date (i, t). The dependent variable vector is therefore defined as:

$$y^{(h)} = \begin{bmatrix} |AR_1^{(h)}| \\ |AR_2^{(h)}| \\ \vdots \\ |AR_N^{(h)}| \end{bmatrix}. \tag{4.2.2}$$

Similarly, the matrix of explanatory variables is constructed as:

$$Z = \begin{bmatrix} z_1^\top \\ z_2^\top \\ \vdots \\ z_N^\top \end{bmatrix}, \tag{4.2.3}$$

where each row corresponds to the reduced-dimensional embedding vector associated with a stock-date observation.

The OLS regression model for the forecasting horizon of h days is given by:

$$y^{(h)} = \beta_0 \mathbf{1} + Z\beta + u, \tag{4.2.4}$$

where β_0 represents the intercept, $\beta \in \mathbb{R}^k$ is the vector of regression coefficients, and $\mathbf{u}$ is the vector of regression errors.

The coefficients are estimated by minimising the sum of squared regression errors:

$$(\hat{\beta}_0, \hat{\beta}) = \arg\min_{\beta_0, \beta} \sum_{j=1}^{N} \left(|AR_j^{(h)}| - \beta_0 - \beta^\top z_j \right)^2. \tag{4.2.5}$$

Thus, for a stock-date m, the predicted absolute abnormal return over a horizon of h days is given by:

$$\widehat{|AR_m^{(h)}|} = \hat{\beta}_0 + \hat{\beta}^\top z_m. \tag{4.2.6}$$

4.3 Experimental Design

This section describes the implementation details specific to the empirical analysis based on the models introduced in Section 4.1 and Section 4.2.

4.3.1 Embedding Aggregation

The models presented in Section 4.1 and Section 4.2 assume a single embedding vector, x_j, for each stock-date observation with an associated price return. However, the dataset contains multiple news articles, and therefore multiple embedding vectors, for the same stock on a given trading day. Therefore, the article-level embeddings must be aggregated to obtain a single embedding for each stock-date observation.

Rather than applying a simple average, the aggregation procedure incorporates the AI-estimated relevance score available in the dataset (see Table 3.1). This allows more

relevant news articles to contribute more strongly to the resulting daily embedding. Given article-level embeddings $x_1, \ldots, x_n$ and their corresponding relevance scores $r_1, \ldots, r_n$ for a given stock-date observation, the aggregated embedding is defined as

$$x_{\text{agg}} = \frac{\sum_{a=1}^{n} r_a x_a}{\sum_{a=1}^{n} r_a}. \tag{4.3.1}$$

This relevance-weighted aggregation produces a single embedding vector for each stock-date observation while preserving the greater influence of articles deemed more relevant. The resulting aggregated embedding, x_{agg}, is subsequently used for the regression in Section 4.2.

A practical consideration is that not all news articles are published on trading days. News may instead be released during weekends or market holidays, when no corresponding stock price observations are available. To ensure that each aggregated embedding is matched with a valid return observation, all news articles published on non-trading days are first assigned to the next available trading day. The aggregation described in Equation 4.3.1 is then performed over all articles assigned to the same stock and trading day. Thus, every aggregated embedding corresponds to a valid stock-date observation with an associated price return.

For brevity, the resulting stock-date embedding is still referred to as the "raw news" embedding or the "pure news" embedding, depending on whether it is constructed from the original or residualised article-level embeddings, rather than explicitly referring to it as the "aggregated" embedding.

4.3.2 Train-Test Split

As discussed in Section 3.3.6, the available "pure news" embeddings span the period from February 2024 to July 2025. The empirical analysis is conducted over the same sample period. Furthermore, the analysis is restricted to the cross-section of the 14 stocks for which both embeddings and corresponding stock price data are available throughout the sample. This results in a dataset consisting of 5,037 stock-date observations over the sample period.

To evaluate the out-of-sample predictive performance of the embeddings, the data is divided into chronologically ordered training and test samples. The training sample consists of observations from February 2024 to March 2025 and the test sample covers the period from April 2025 to July 2025. This chronological split is primarily to avoid any look-ahead bias and therefore provides a realistic evaluation of the predictive performance. The same train-test split is used throughout all empirical analyses presented in this chapter, unless otherwise stated.

4.3.3 Absolute Abnormal Returns Construction

Equation 4.1.2 is used to construct abnormal returns over the forecasting horizons $h \in \{1, 5, 21\}$ trading days. The market return in Equation 4.1.1 is measured using the S&P 500 Index. For each stock i and each forecasting horizon h, the parameters, $\hat{\alpha}_i^{(h)}$ and $\hat{\beta}_i^{(h)}$, are estimated using only the training sample. The resulting coefficient estimates are then used in Equation 4.1.2 to obtain the abnormal returns for the training sample, after which their absolute values are taken to form the prediction target. To construct the prediction target for the test sample, the same estimated training-sample parameters, $\hat{\alpha}_i^{(h)}$ and $\hat{\beta}_i^{(h)}$, are used in Equation 4.1.2.

4.3.4 Estimation of Parameters

The absolute abnormal returns of the training sample are used to construct the dependent variable vector $\mathbf{y}^{(h)}$ defined in Equation 4.2.2. The embedding vectors are transformed using Equation 4.2.1 to obtain the reduced-dimensional representations $z_{i,t}$, which are subsequently used to construct the explanatory variable matrix $\mathbf{Z}$ in Equation 4.2.3. The PCA transformation is estimated using the training sample and the same transformation is then applied to the test sample embeddings to obtain their corresponding reduced-dimensional representations. The OLS model specified in Equation 4.2.4 is then estimated on the training sample, with the coefficients obtained by minimising the objective function in Equation 4.2.5. The resulting estimates are finally used in Equation 4.2.6 to generate predictions of absolute abnormal returns for the test sample.

4.3.5 Evaluation Criterion

The predictive performance of the regression framework, for a given number of principal components k, is evaluated using the coefficient of determination, R^2. Formally, the out-of-sample R^2 is defined as:

$$R^2_k = 1 - \frac{\sum_{j=1}^{N_{\text{test}}} \left(|AR_j^{(h)}| - \widehat{|AR_j^{(h)}|} \right)^2}{\sum_{j=1}^{N_{\text{test}}} \left(|AR_j^{(h)}| - \overline{|AR^{(h)}|} \right)^2},$$

where N_{test} denotes the number of observations in the test sample, $|AR_j^{(h)}|$ represents the observed absolute abnormal return obtained from Equation 4.1.2, $\widehat{|AR_j^{(h)}|}$ represents the predicted absolute abnormal return obtained from Equation 4.2.6, and $\overline{|AR^{(h)}|}$ denotes the mean absolute abnormal return in the test sample.

The choice of the number of principal components, k, represents an important modelling decision, as different levels of dimensionality reduction may lead to different predictive performances. A smaller number of components may remove information contained in the original embedding space, while a larger number of components may increase the risk of overfitting, particularly given the relatively limited number of stock-date observations available in the sample. Therefore, rather than selecting a single value of k, the regression framework is estimated for each number of principal components ranging from $k = 1$ to $k = 500$. The upper limit of 500 components is chosen as a compromise between retaining sufficient information from the original 3,072-dimensional embedding space and reducing the risk of overfitting.

The resulting sequence of out-of-sample R^2 values provides a measure of predictive performance as a function of embedding dimensionality. To summarise the predictive performance across different dimensionality choices, the area under the curve (AUC) of the R^2 profile is calculated. The AUC captures the overall predictive performance across the range of principal component specifications rather than relying on a single arbitrary choice of k. It is defined as:

$$\text{AUC} = \int_1^{500} R^2_k \, dk.$$

In practice, since the R^2 values are evaluated at discrete values of k, the integral is approximated using the trapezoidal rule:

$$\text{AUC} = \sum_{k=1}^{499} \frac{R^2_k + R^2_{k+1}}{2}.$$

The AUC is therefore used as the primary evaluation metric, as it captures the predictive performance of the embeddings across a broad range of dimensionality choices and avoids relying on a single potentially arbitrary PCA specification, and is used to compare the predictive performance of "raw news" and "pure news" embeddings.

4.4 Empirical Evaluation

This section presents the empirical evaluation of the predictive performance of the news embeddings using the models presented in Section 4.1 and Section 4.2 following the experimental design formed in Section 4.3.

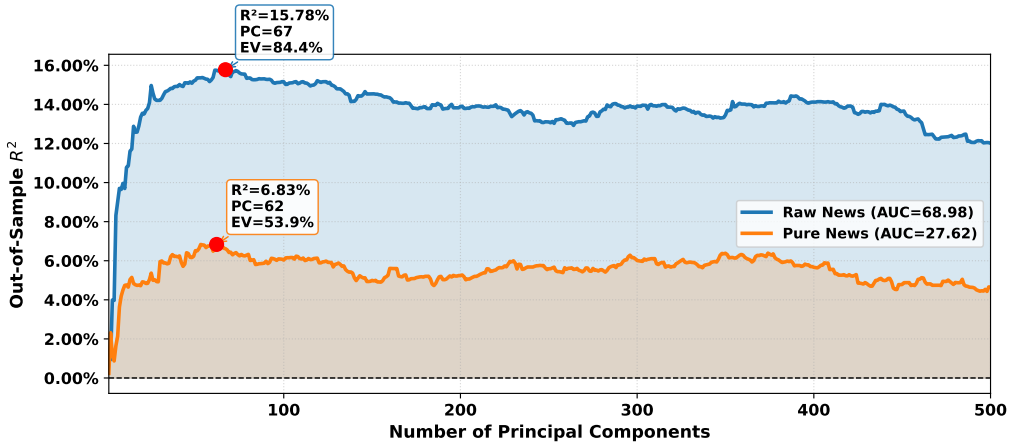
Figure 4.1 presents the AUC of the "raw news" and "pure news" embeddings across the 1-day, 5-day, and 21-day horizons. Each panel also reports the principal component specification at which the highest out-of-sample R^2 is achieved, together with the proportion of variance in the embedding space explained by the corresponding number of principal components.

The results indicate that the predictive performance of the "raw news" and "pure news" embeddings differs substantially across forecasting horizons. At the 1-day horizon, "raw news" exhibits considerably stronger predictive performance, with an AUC of 68.98 and a maximum out-of-sample R^2 of 15.78% achieved using 67 principal components. In comparison, "pure news" obtains a substantially lower AUC of 27.62, with its highest out-of-sample R^2 of 6.83% achieved using 62 principal components. Moreover, "raw news" consistently maintains higher R^2 values than "pure news" rather than converging as the number of components increases.

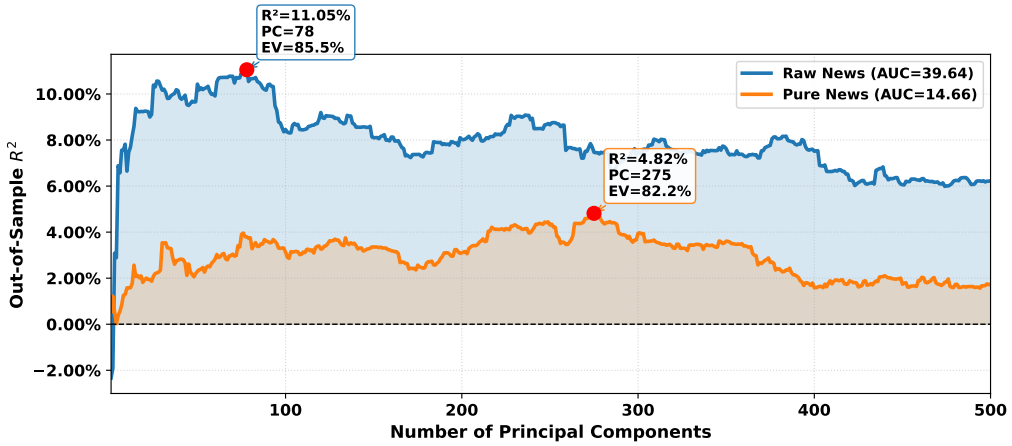
A similar pattern is observed for the 5-day forecasting horizon as well, although the overall predictive performance is lower than at the 1-day horizon. The AUC for "raw news" is 39.64, compared with 14.66 for "pure news". The highest out-of-sample R^2 for "raw news" is 11.05%, achieved at 78 principal components, whereas "pure news" achieves a maximum of 4.82% at 275 principal components. As in the 1-day case, "raw news" consistently outperforms "pure news" across the range of principal component specifications, with no clear evidence of convergence.

The results change considerably for the 21-day forecasting horizon. Both representations exhibit substantially weaker predictive performance, with the "raw news" embedding producing an AUC of -28.47 and failing to achieve a positive out-of-sample R^2 for any number of principal components. The "pure news" embedding performs somewhat better in relative terms, with an AUC of -11.13 and a maximum out-of-sample R^2 of only 0.75%, achieved at 134 principal components. Although the predictive performance remains weak for both representations, "pure news" consistently produces higher R^2 values than "raw news".

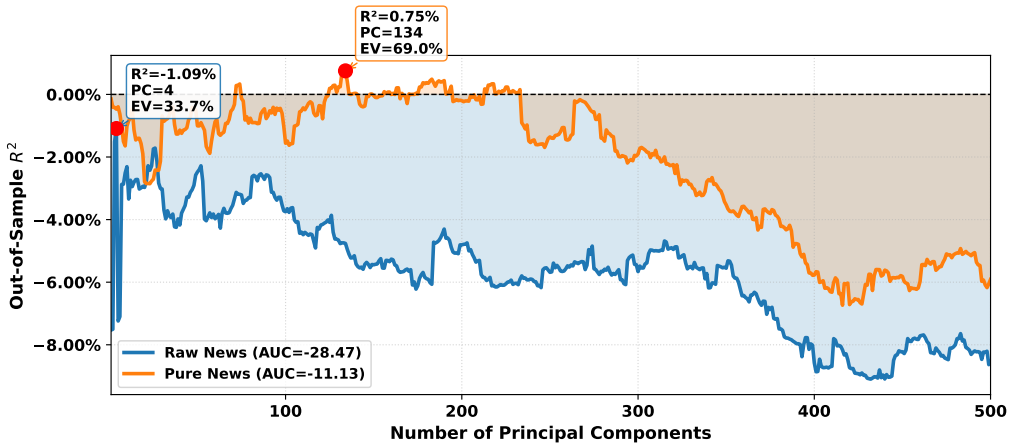
One possible interpretation of these results is that, when predicting short-term price movements, information about persistent differences between firms contained in the news representations may provide useful context for interpreting the news observed on a given day. Consider, for example, a day on which several news articles are published about *NVIDIA*, covering different topics and originating from different sources. Even when the articles themselves contain different information, they are all associated with a company whose characteristics and typical market behaviour provide a persistent context for interpreting the news. Similarly, news concerning *Apple* may be interpreted differently from otherwise comparable news concerning *NVIDIA* or *Microsoft*, because the historical behaviour and characteristics of these firms differ. Thus, when using the collection of news articles observed for a stock on a particular day to predict its subsequent price movement, retaining this persistent firm-level information may allow the model to make more accurate short-horizon predictions than when this information has been removed. This



(a) 1-day forecasting horizon



(b) 5-day forecasting horizon



(c) 21-day forecasting horizon

Figure 4.1: Out-of-sample R^2 profiles and AUC evaluation for "raw news" and "pure news" embeddings. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.

interpretation is also consistent with the weaker predictive performance observed at the 21-day horizon. If the same news concerning, for example, *Apple* is used to predict its price movement one month later, the information contained in the articles may have become substantially less informative because many other pieces of information have arrived in the intervening period. In this setting, the persistent firm-level information contained in the representation is also less likely to provide sufficient information to accurately predict the eventual price movement. The results may therefore suggest that persistent firm-level context is particularly useful when interpreting news for short-horizon price movements, while its usefulness diminishes as the forecasting horizon increases. The potential factors contributing to the differences in predictive performance between "raw news" and "pure news" are examined further in Section 4.6.

4.5 Robustness Checks

In this section, the analysis is next extended to different groups of news available within the dataset. The same AUC-based evaluation is applied separately to each news source type and news category. This analysis serves two purposes. First, it examines whether the predictive performance observed in the full sample is driven primarily by particular groups of news articles, which may help identify whether specific groups contribute disproportionately to the stronger or weaker performance observed across forecasting horizons. Second, it allows the contribution of "raw news" and "pure news" embeddings to the prediction of absolute abnormal returns to be compared across different groups of articles.

4.5.1 Predictive Performance Across News Source Types

We first examine whether the predictive performance differs across news source types. News source type is one of the variables available in the dataset (see Table 3.1). For each source type, articles are grouped according to their assigned source category, and the corresponding embeddings are used to evaluate how much variation in absolute abnormal returns can be predicted from news originating from that source.

The regression framework and experimental design remain the same as those described in Sections 4.1, 4.2, and 4.3. The only difference is that, rather than using the full set of available news articles, the analysis is conducted separately for each source type. For example, when evaluating financial media, only articles classified as financial media are retained. These articles are then aggregated at the stock-date level using the relevance-weighted aggregation procedure described in Section 4.3.1. If there is no embedding for financial media for a particular stock-date, then the zero-vector is considered as the stock-date embedding. The chronological train-test split described in Section 4.3.2 is maintained. The construction of absolute abnormal returns in Section 4.3.3 remains unchanged, as it depends only on the stock and market returns and is therefore independent of the news embeddings. For each source type, PCA is then fitted to the corresponding aggregated embeddings using the training sample, and the resulting principal components are used in the OLS regression described in Section 4.3.4. The evaluation procedure, including the calculation of out-of-sample R^2 and AUC across the range of principal component specifications, remains identical to that described in Section 4.3.5.

To ensure that the results are based on a sufficiently representative set of observations, the analysis is restricted to source types for which at least 70% of the stock-date observations have coverage after the aggregation step. Thus, the regression analysis is performed for five news source types: Financial Media, Mainstream, Research, Social Media, and Trade Media.

Since reporting three R^2 profile plots for each of the five news source types would result

in a large number of figures, for compact comparison, Table 4.1 only reports the AUC, the highest out-of-sample R^2, the corresponding number of principal components (k), and the proportion of explained variance (EV) at that specification for both "raw news" and "pure news" embeddings across the 1-day, 5-day, and 21-day forecasting horizons. To see the complete R^2 profiles for each news source type and forecasting horizon, see Appendix C.1.

Table 4.1: Predictive performance of "raw news" and "pure news" embeddings across news source types. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.

(a) 1-day forecasting horizon

	Raw News				Pure News			
News Source	AUC	Best R^2	k	EV (%)	AUC	Best R^2	k	EV (%)
Fin. Media	75.89	16.95	139	86.02	44.68	11.11	418	89.22
Mainstream	40.53	11.64	27	66.39	14.34	4.50	151	70.96
Research	55.11	13.86	105	81.26	12.09	3.96	140	71.45
Social Media	60.01	13.36	205	86.00	29.08	7.38	245	77.03
Trade Media	37.66	10.52	34	59.77	17.62	6.38	82	53.04
All Sources	**68.98**	**15.78**	**67**	**84.40**	**27.62**	**6.83**	**62**	**53.90**

(b) 5-day forecasting horizon

	Raw News				Pure News			
News Source	AUC	Best R^2	k	EV (%)	AUC	Best R^2	k	EV (%)
Fin. Media	43.49	10.88	51	76.26	33.00	8.63	230	80.12
Mainstream	34.64	8.44	67	76.35	11.96	4.48	254	80.62
Research	31.15	9.85	17	58.22	10.05	3.00	352	87.17
Social Media	23.81	8.29	6	34.29	14.32	4.23	239	76.56
Trade Media	16.73	7.70	8	36.61	7.59	5.99	121	61.10
All Sources	**39.64**	**11.05**	**78**	**85.50**	**14.66**	**4.82**	**275**	**82.20**

(c) 21-day forecasting horizon

	Raw News				Pure News			
News Source	AUC	Best R^2	k	EV (%)	AUC	Best R^2	k	EV (%)
Fin. Media	-18.94	0.66	10	50.76	0.91	2.31	105	65.57
Mainstream	-25.54	0.98	5	32.48	-5.16	1.80	65	54.66
Research	-19.82	-0.13	6	37.03	-6.23	2.73	30	42.49
Social Media	-33.11	0.49	6	34.29	-21.63	-0.30	1	6.65
Trade Media	-27.97	2.57	3	20.73	-9.12	4.73	92	55.38
All Sources	**-28.47**	**-1.09**	**4**	**33.70**	**-11.13**	**0.75**	**134**	**69.00**

At the 1-day forecasting horizon, all five news source types exhibit positive AUC values for both "raw news" and "pure news", indicating positive predictive performance across the different source groups. However, "raw news" consistently outperforms "pure news" for every source type. Financial Media achieves the strongest performance for both representations, with an AUC of 75.89 and the highest out-of-sample R^2 of 16.95% for "raw news", compared with an AUC of 44.68 and the highest out-of-sample R^2 of 11.11% for "pure news". Social Media provides the second-highest AUC for both "raw news" and "pure news". Although the AUC generally decreases when moving from "raw news" to

"pure news", the reduction is particularly pronounced for Research news. For this source type, the AUC falls from 55.11 to 12.09, while the highest out-of-sample R^2 decreases from 13.86% to 3.96%. Moreover, Financial Media achieves an AUC above that obtained when all news sources are included, as reported in Section 4.4, for both "raw news" and "pure news". For "raw news", it is the only source type to exceed the corresponding full-sample AUC, whereas for "pure news", Social Media also achieves an AUC above the full-sample result.

At the 5-day forecasting horizon, the AUC remains positive for both "raw news" and "pure news" across all five source types, although the overall AUC values for "raw news" are lower than those observed at the 1-day horizon. The decrease is less pronounced for "pure news", whose AUC values remain relatively closer to those at the 1-day horizon. Financial Media continues to exhibit the highest AUC for both representations, with values of 43.49 for "raw news" and 33.00 for "pure news". The ordering of the other source types differs somewhat between the two representations. For "raw news", Mainstream has the second-highest AUC at 34.64, followed by Research and Social Media, whereas for "pure news", Social Media has the second-highest AUC at 14.32. Furthermore, Financial Media is the only source type to exceed the corresponding full-sample AUC for both representations, with 43.49 exceeding 39.64 for "raw news" and 33.00 exceeding 14.66 for "pure news".

At the 21-day forecasting horizon, the results are considerably weaker for both representations. The AUC for "raw news" is negative for all five source types, while "pure news" also produces negative AUC values for four of the five source types and a value close to zero for Financial Media (0.91). In this horizon, "pure news" has a higher AUC than "raw news" for every source type. The highest out-of-sample R^2 values for "raw news" are positive for most source types, but are generally small. The largest is observed for Trade Media at 2.57%, although this occurs at only three principal components. The corresponding results for "pure news" are similarly small, with the highest value being 4.73% for Trade Media. Relative to the full-sample results, Financial Media, Mainstream, Research, and Trade Media all achieve higher AUCs for both representations, while Social Media remains below the full-sample AUC for both "raw news" and "pure news".

One possible interpretation of the differences across news source types is that the predictive value of news depends on the type of information typically conveyed by each source. Financial Media consistently produces the strongest results at the 1-day and 5-day horizons for both "raw news" and "pure news", which may reflect its focus on information that is directly relevant to financial markets, such as earnings, valuation, corporate developments, and changes in market expectations. Social Media also performs relatively strongly at the 1-day horizon for "raw news", potentially because information disseminated through these channels can be closely connected to events or reactions that are reflected in prices over short time horizons. However, its performance falls considerably for "pure news", which may indicate that removing persistent firm-level differences removes information that is useful for interpreting this type of news. Research exhibits a related pattern. It performs relatively well for "raw news" at the 1-day horizon, but its predictive performance declines substantially for "pure news". This may be because research-oriented articles often contain analysis that is meaningful in conjunction with the broader context of the firms being discussed. Mainstream Media produces more moderate performance, which may reflect its broader coverage of events that are not always directly related to information that translates into short-term stock-price movements. Trade Media also performs relatively weakly, although its specialised coverage may still contain information relevant to particular firms or events. At the 21-day horizon, the predictive performance of nearly all source types becomes weak for both representations, suggesting that the information contained in individual source types is more closely related to shorter-term

price movements than to movements over longer horizons. Although these interpretations provide possible explanations for the observed differences, the results do not allow the specific mechanisms underlying the source-level differences to be established.

The fact that some source types outperform the full-sample AUC, while others do not, may reflect a trade-off between reducing heterogeneity and retaining useful information. When articles from all source types are combined, the stock-date embedding incorporates information from sources that may differ substantially in both the type and perspective of the information they provide. This heterogeneity may make the resulting representation less focused and potentially introduce noise into the predictive signal. Restricting the aggregation to a particular source type can reduce this heterogeneity and produce a more homogeneous representation, which may make information relevant for subsequent price movements more pronounced and easier for the regression model to capture. This provides one possible explanation for why Financial Media exceeds the full-sample AUC at the 1-day and 5-day horizons, and why some other source-specific results also exceed the full-sample AUC at certain horizons. However, restricting the sample to a single source type also removes information that may be useful and complementary to the information contained in that source. Thus, a source may become more focused without becoming more predictive if the information discarded from the other sources is sufficiently valuable. This may explain why several source types remain below the full-sample AUC at the shorter horizons. At the 21-day horizon, several source types exceed the full-sample AUC, but this should be interpreted cautiously because the full-sample AUC is itself negative. A higher AUC in this setting indicates relatively better performance rather than strong predictive ability.

Taking everything into account, the source-specific results show a pattern that is broadly consistent with the results obtained for the full sample in Section 4.4. At the 1-day and 5-day horizons, "raw news" produces positive AUC values and consistently outperforms "pure news" across the different source types. At the 21-day horizon, the predictive performance of both representations is substantially weaker, with "pure news" performing slightly better than "raw news" across all source types. The results therefore show that the patterns observed in the full sample are also present when the analysis is conducted separately by news source type.

4.5.2 Predictive Performance Across News Tags

The analysis can next be extended from news source types to the news tags assigned to each article. Since each article in the dataset is assigned two news tags (see Table 3.1), the articles are grouped according to each of their assigned tags, and the corresponding embeddings are used to evaluate the variation in absolute abnormal returns that can be explained by each news category. The same regression framework and experimental design used for the news source analysis are maintained. However, because each article has two assigned tags, a given article may contribute to two separate tag-specific regressions. Thus, the tag-specific samples are not mutually exclusive, and the same article can be represented in the embedding aggregation for more than one news tag.

As in the news source analysis, the tag-specific regressions are restricted to news tags for which at least 70% of the stock-date observations have coverage after the aggregation. Based on this criterion, the analysis is performed for eleven news tags: Valuation, Strategy, Outlook, Earnings, Competition, Operations, Momentum, Launches, Legal Regulatory, Analyst Ratings, and Macro.

For consistency with the news source analysis, the complete R^2 profiles for each news tag and forecasting horizon are provided in Appendix C.2 and Table 4.2 reports the AUC, the highest out-of-sample R^2, the corresponding number of principal components (k), and the proportion of explained variance (EV) at that specification for both "raw news" and

"pure news" embeddings across the 1-day, 5-day, and 21-day forecasting horizons.

The results at the 1-day forecasting horizon show that the AUC is positive for both "raw news" and "pure news" for the majority of news tags. Valuation exhibits the strongest predictive performance for both representations, achieving an AUC of 77.33 for "raw news" and 58.95 for "pure news", as well as the highest out-of-sample R^2 among the tags for each representation. Beyond Valuation, however, the relative ranking of the news tags differs between "raw news" and "pure news". For most tags, "raw news" produces higher AUC values than "pure news". The difference is particularly pronounced for Strategy, where the AUC falls from 36.67 for "raw news" to only 0.02 for "pure news". There are, however, some exceptions to the general dominance of "raw news". For Momentum, the AUCs of "raw news" and "pure news" are relatively close, at 35.07 and 33.13, respectively. More notably, "pure news" performs better than "raw news" for Macro, with an AUC of 44.71 compared with 29.41. Relative to the full-sample results, Valuation is the only tag whose "raw news" AUC exceeds the full-sample AUC of 68.98, whereas for "pure news", Valuation, Earnings, Momentum, and Macro all exceed the full-sample AUC of 27.62. Thus, although the general pattern at the 1-day horizon is consistent with the full-sample and the news source results, the tag-level analysis also indicates that the relative performance of the two representations varies across news categories.

At the 5-day forecasting horizon, the AUC remains positive for both representations across most news tags. The main exceptions are Launches and Legal Regulatory for "pure news", which produce slightly negative AUC values of -0.08 and -5.23, respectively. For "raw news", Earnings produces the highest AUC at 32.68, slightly exceeding Valuation at 32.03. For "pure news", in contrast, Valuation remains the strongest-performing category, with an AUC of 30.03. The results also contain some differences from the 1-day horizon such as the Strategy tag performing better for "pure news" at the 5-day horizon, increasing from an AUC of 0.02 at 1-day to 11.83 at 5-day. The difference between "raw news" and "pure news" is also generally smaller at the 5-day horizon than at the 1-day horizon. For example, the AUCs for Valuation are 32.03 and 30.03, respectively, while the corresponding differences for Competition, Momentum, and Macro are also relatively small. This indicates that the stronger performance of "raw news" observed at the 1-day horizon is less pronounced over a 5-day horizon, although "raw news" continues to perform better for most tags. Furthermore, relative to the full-sample results, none of the individual tags exceeds the full-sample AUC of 39.64 for "raw news". For "pure news", however, Valuation, Outlook, Earnings, Competition, Momentum, and Analyst Ratings all exceed the full-sample AUC of 14.66.

The results change more substantially at the 21-day forecasting horizon. AUC values are negative for "raw news" across all eleven tags, while "pure news" generally produces less negative AUC values and performs better than "raw news" for every tag. The difference is particularly notable for Valuation. While "raw news" produces an AUC of -24.70 and a maximum out-of-sample R^2 of -2.94%, "pure news" produces an AUC of 12.54 and a maximum out-of-sample R^2 of 4.94%. This makes Valuation the clearest example in which the removal of firm-characteristic information is associated with stronger predictive performance at the longer horizon. Several other tags also produce less negative AUC values for "pure news", although the overall predictive performance at this horizon remains weak. Moreover, Valuation, Earnings, Operations, and Legal Regulatory produce higher AUCs than the full-sample AUC of -28.47 for "raw news". For "pure news", however, Valuation, Strategy, Outlook, Earnings, Operations, and Momentum exceed the full-sample AUC of -11.13.

One possible interpretation of the differences across news tags is that their predictive value depends on the type of information conveyed. Valuation and Earnings generally produce stronger short-horizon results, which may reflect the direct connection between these

Table 4.2: Predictive performance of "raw news" and "pure news" embeddings across news tags. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.

(a) 1-Day Horizon

	Raw News				Pure News			
News Tag	AUC	Best R^2	k	EV (%)	AUC	Best R^2	k	EV (%)
Valuation	77.33	16.46	310	94.03	58.95	13.44	207	81.09
Strategy	36.67	9.87	17	55.06	0.02	2.12	84	55.19
Outlook	46.82	12.54	21	62.98	19.73	6.03	162	72.94
Earnings	47.45	11.08	124	85.40	28.98	7.45	340	86.96
Competition	33.44	10.56	96	79.19	20.11	6.43	65	55.86
Operations	35.01	10.25	35	58.88	15.62	5.23	170	68.68
Momentum	35.07	10.78	55	76.68	33.13	9.78	117	72.11
Launches	29.17	9.84	22	55.68	18.62	5.30	79	53.56
Legal Regulatory	38.25	10.93	64	71.54	26.37	7.73	225	78.38
Analyst Ratings	30.93	10.76	19	67.90	13.84	4.68	140	73.81
Macro	29.41	8.28	102	79.40	44.71	12.34	174	79.56
All Tags	**68.98**	**15.78**	**67**	**84.40**	**27.62**	**6.83**	**62**	**53.90**

(b) 5-Day Horizon

	Raw News				Pure News			
News Tag	AUC	Best R^2	k	EV (%)	AUC	Best R^2	k	EV (%)
Valuation	32.03	9.62	32	75.54	30.03	8.18	55	58.32
Strategy	27.58	8.94	11	46.62	11.83	4.16	213	74.31
Outlook	28.48	9.18	42	71.40	18.51	5.42	308	84.09
Earnings	32.68	10.75	90	82.47	25.47	7.62	147	73.55
Competition	18.99	8.24	5	36.22	17.30	5.77	192	76.53
Operations	17.17	8.63	12	42.52	7.89	4.00	145	65.30
Momentum	26.08	8.49	6	42.04	27.91	8.94	108	70.71
Launches	8.10	8.05	7	37.56	-0.08	4.21	32	36.63
Legal Regulatory	16.67	8.91	8	41.27	-5.23	3.22	59	51.88
Analyst Ratings	28.18	8.64	39	76.28	21.17	6.75	195	79.44
Macro	10.28	8.04	100	79.16	10.27	8.88	48	57.72
All Tags	**39.64**	**11.05**	**78**	**85.50**	**14.66**	**4.82**	**275**	**82.20**

(c) 21-Day Horizon

	Raw News				Pure News			
News Tag	AUC	Best R^2	k	EV (%)	AUC	Best R^2	k	EV (%)
Valuation	-24.70	-2.94	24	72.19	12.54	4.94	170	77.99
Strategy	-33.44	-0.02	8	39.42	-4.30	2.18	31	36.58
Outlook	-35.08	0.20	11	50.34	0.26	1.58	40	47.35
Earnings	-28.40	-0.88	1	16.24	-6.72	4.76	73	61.03
Competition	-60.26	0.02	10	49.67	-40.80	-0.23	1	5.04
Operations	-22.65	1.97	12	42.52	-0.67	3.90	74	51.27
Momentum	-38.30	2.60	6	42.04	-6.51	1.17	74	64.01
Launches	-75.39	-1.24	1	14.14	-65.75	-0.65	1	6.39
Legal Regulatory	-27.89	1.74	54	69.41	-12.62	6.65	117	65.55
Analyst Ratings	-43.26	-0.71	1	25.79	-29.81	1.46	28	44.48
Macro	-48.98	3.99	17	56.52	-45.71	2.89	16	37.06
All Tags	**-28.47**	**-1.09**	**4**	**33.70**	**-11.13**	**0.75**	**134**	**69.00**

types of news and investors' assessment of a firm's value and future performance. Outlook and Momentum also perform relatively well, potentially because changes in expectations about a firm's prospects or recent price behaviour can be reflected relatively quickly in subsequent price movements. In contrast, categories such as Launches, Operations, and Analyst Ratings generally produce weaker results, possibly because the information they contain is either more difficult to translate into an immediate price response or depends more heavily on additional context. The results for Strategy provide an interesting example of this distinction. Its "pure news" AUC is only 0.02 at the 1-day horizon, compared with 36.67 for "raw news", but increases to 11.83 at the 5-day horizon. One possible explanation is that strategic developments are more difficult to interpret in isolation immediately after publication, since their implications depend on the firm's existing position, business model, and longer-term objectives. These persistent firm-level differences are retained in "raw news" and may therefore provide useful context for the immediate price response, whereas their removal in "pure news" may leave insufficient information to identify that response. As the horizon extends, however, investors may have more time to assess the implications of strategic information, allowing the information remaining in "pure news" to become more useful.

Macro provides a contrasting case. At the 1-day horizon, "pure news" achieves an AUC of 44.71, substantially higher than the 29.41 obtained for "raw news", while the two representations produce almost identical AUCs at the 5-day horizon. One possible explanation is that macroeconomic news is inherently less dependent on persistent firm-level differences, since the information concerns economic conditions that can affect firms more broadly. Removing these differences may therefore leave a relatively clear common signal, potentially explaining the stronger short-horizon performance of "pure news". Over five days, however, the additional firm-level context retained in "raw news" appears to provide little incremental information. At the 21-day horizon, Valuation provides another notable case, with "pure news" achieving a positive AUC of 12.54 despite the removal of persistent firm-level differences. A possible explanation is that valuation-related news can capture changes in how investors assess a firm's future earnings, growth prospects, or risks, and these changes may take longer to be incorporated into prices. In this setting, removing persistent firm-level differences may help isolate changes in valuation rather than information about the firm's characteristics that has remained relatively stable over time. This could explain why "pure news" retains predictive information at the 21-day horizon while "raw news" performs poorly.

The differences relative to the full-sample AUC may partly arise from how the news is aggregated. In the full-sample analysis, articles covering different news tags are combined into a single stock-date embedding. For "pure news", persistent firm-level differences have first been removed from the articles, so the remaining representations are intended to capture information beyond these persistent characteristics. Aggregating these residualised representations across different types of news may produce a more heterogeneous and potentially noisier daily representation, as information from unrelated topics is combined into a single vector. When the aggregation is instead performed separately for a particular news tag, the resulting embedding is more focused on a specific type of information, which may make the relevant predictive signal more pronounced. This provides one possible explanation for why several tags exceed the full-sample AUC for "pure news". At the 1-day horizon, Valuation, Earnings, Momentum, and Macro exceed the full-sample AUC of 27.62, while at the 5-day horizon, Valuation, Outlook, Earnings, Competition, Momentum, and Analyst Ratings exceed the full-sample AUC of 14.66. For "raw news", in contrast, only Valuation exceeds the full-sample AUC at the 1-day horizon, while no individual tag exceeds it at the 5-day horizon. This difference may suggest that the reduction in heterogeneity from tag-specific aggregation is particularly useful when the

persistent firm-level differences have already been removed from the news representations. However, focusing on a single tag also discards potentially useful information contained in other categories, so a more homogeneous representation does not necessarily imply better predictive performance. At the 21-day horizon, several tags exceed the full-sample AUC for both representations, but these comparisons should be interpreted cautiously because the full-sample AUC is negative, meaning that a higher AUC represents relatively better rather than strong predictive performance.

Hence, the results broadly reproduce the patterns observed in the full-sample and news source analysis while also revealing some heterogeneity across news categories. At the 1-day and 5-day horizons, AUC values are generally positive, with "raw news" outperforming "pure news" for most tags. However, this pattern is not universal. Macro at the 1-day horizon and several other categories show relatively similar or stronger performance for "pure news". At the 21-day horizon, the relationship reverses, with "raw news" producing negative AUC values across all tags and "pure news" generally performing less poorly. Valuation is a notable exception to the generally weak long-horizon performance, as its "pure news" representation achieves a positive AUC and a maximum out-of-sample R^2 of 4.94%. Thus the results suggest that the relative predictive performance of "raw news" and "pure news" varies across both news categories and forecasting horizons, while the broad horizon-dependent pattern remains consistent with that observed in the full sample.

4.6 Investigating the Role of Firm Identity

The results presented in Section 4.4 and Section 4.5 show that the relative predictive performance of "raw news" and "pure news" embeddings varies across forecasting horizons, news source types, and news categories. A natural question is therefore what might account for the difference in predictive performance between the two representations. Identifying the precise reason is non-trivial, as the difference between the embeddings may arise from properties of their geometry or from information encoded in the embedding space that is not directly observable from the regression results. A detailed analysis of these geometric properties is beyond the scope of this study. Instead, this section provides an empirical investigation of one potential explanation for the observed differences.

As discussed in Section 3.3.6, the "pure news" embeddings are constructed by removing information associated with firm characteristics from the original embedding representation. The residualisation procedure is therefore intended to reduce the extent to which the embeddings encode persistent firm-specific information, while retaining information that cannot be explained by these characteristics. This raises the question of how much information about firm identity remains encoded in each representation after this procedure.

To investigate this, a multi-class classification framework is used to assess whether the firm associated with a given embedding can be identified from the embedding representation. The underlying intuition is that if an embedding contains substantial information about firm identity, a classifier should be able to distinguish between firms using the embedding as input. Hence, weaker classification performance would suggest that less firm-specific information is encoded in the representation. The classification analysis is applied to both "raw news" and "pure news" embeddings and is subsequently extended to the embeddings constructed from different news source types and news categories. Comparing the classification performance of these representations with their corresponding AUCs across different forecasting horizons provides an empirical basis for whether the amount of firm-specific information encoded in the embeddings is associated with their predictive performance.

We first briefly describe the multi-class classification framework employed in this study, followed by the experimental design choices, such as the train-test split and classification

evaluation criterion, before discussing the empirical results.

4.6.1 Multi-Class Classification Framework

The classification framework is based on multinomial logistic regression. Let $x_j \in \mathbb{R}^d$ denote the embedding vector for observation j, and let $y_j \in \{1, \ldots, C\}$ denote its associated firm, where C is the total number of firms. For each firm $c \in \{1, \ldots, C\}$, the multinomial logistic regression model is defined by

$$P(y_j = c \mid x_j) = \frac{\exp\left(\beta_{0,c} + \beta_c^\top x_j\right)}{\sum_{l=1}^{C} \exp\left(\beta_{0,l} + \beta_l^\top x_j\right)}, \tag{4.6.1}$$

where $P(y_j = c \mid x_j)$ denotes the conditional probability that observation j belongs to firm c given its embedding vector x_j, the function $\exp(\cdot)$ denotes the exponential function, the vector $\beta_c \in \mathbb{R}^d$ is the coefficient vector associated with firm c, and $\beta_{0,c} \in \mathbb{R}$ is its corresponding intercept.

The parameters of the model are estimated by minimising the multinomial cross-entropy loss over the training observations. Let $\mathbf{1}_{\{y_j=c\}}$ denote an indicator function that takes the value 1 if observation j belongs to firm c and 0 otherwise. The loss function is given by

$$\mathcal{L}(\boldsymbol{\beta}) = -\frac{1}{N_{\text{train}}} \sum_{j=1}^{N_{\text{train}}} \sum_{c=1}^{C} \mathbf{1}_{\{y_j=c\}} \log P(y_j = c \mid x_j),$$

where N_{train} denotes the number of observations in the training sample and $\boldsymbol{\beta}$ denotes the collection of all coefficient vectors and intercepts. The estimated parameters are therefore obtained as

$$\left(\hat{\boldsymbol{\beta}}_0, \hat{\boldsymbol{\beta}}\right) = \arg\min_{\boldsymbol{\beta}_0, \boldsymbol{\beta}} \mathcal{L}(\boldsymbol{\beta}),$$

where $\hat{\boldsymbol{\beta}}_0$ and $\hat{\boldsymbol{\beta}}$ denote the estimated intercepts and coefficient vectors, respectively.

Given the estimated parameters, the predicted probability of each firm can be obtained by substituting the estimated parameters into Equation 4.6.1, i.e., define the linear score for firm c as

$$s_{jc} = \hat{\beta}_{0,c} + \hat{\beta}_c^\top x_j.$$

The predicted probability of firm c can then be written as

$$\hat{P}(y_j = c \mid x_j) = \frac{\exp(s_{jc})}{\sum_{l=1}^{C} \exp(s_{jl})}.$$

The predicted firm is subsequently obtained by selecting the class with the highest predicted probability:

$$\hat{y}_j = \arg\max_{c \in \{1,\ldots,C\}} \hat{P}(y_j = c \mid x_j).$$

4.6.2 Train-Test Split

To train the classifier, the same train-test split mentioned in Section 4.3.2 is maintained. Observations from February 2024 to March 2025 are used to train the classifier, while observations from April 2025 to July 2025 are reserved for out-of-sample evaluation. We maintain the same chronological split to ensure consistency with the regression framework

and, more importantly, to avoid potential look-ahead bias when comparing classification performance with the predictive performance of the embeddings. It is also important to note that, unlike the regression framework, no aggregation is performed prior to classification. The classifier is used solely to assess how effectively the article-level embeddings can be used to predict the associated firm within the specified train-test window and therefore does not require observations to be unique at the stock-date level.

4.6.3 Evaluation Criterion

The primary evaluation metric for the classifier is balanced accuracy. This metric is used instead of conventional accuracy because the number of observations may differ across firms, and therefore a classifier could obtain a relatively high overall accuracy by performing well primarily on firms with a larger number of observations. Balanced accuracy mitigates this issue by assigning equal weight to each firm when calculating the overall classification performance.

For each firm c, let TP_c denote the number of observations correctly classified as firm c, and let N_c denote the total number of observations in the test sample belonging to firm c. The recall, or true positive rate, for firm c is therefore

$$\text{Recall}_c = \frac{TP_c}{N_c}.$$

Balanced accuracy is defined as the arithmetic mean of the recall values across all C firms:

$$\text{Balanced Accuracy} = \frac{1}{C} \sum_{c=1}^{C} \text{Recall}_c.$$

Hence, each firm contributes equally to the final metric regardless of the number of observations associated with that firm. A balanced accuracy of 1 corresponds to perfect classification, while a value of $1/C$ corresponds to the expected performance of a classifier that randomly assigns observations uniformly across the C firms. Higher balanced accuracy, therefore, indicates that the embedding contains more information that allows the associated firm to be distinguished from the other firms in the sample.

4.6.4 Empirical Evaluation

First, we train the classifier described in Section 4.6.1 using the train-test split described in Section 4.6.2 for both the "raw news" and "pure news" embeddings. The classification is performed using all available news articles, without restricting the sample to any particular news source type or category. The resulting balanced accuracy is 83.51% for the "raw news" embeddings and 24.34% for the "pure news" embeddings. The substantial reduction in balanced accuracy indicates that the "pure news" embeddings contain considerably less information that can be used to distinguish between firms. This provides empirical evidence that the residualisation procedure described in Section 3.3.6 successfully removes a substantial component of the firm-specific information encoded in the original embeddings.

Comparing these results with the corresponding AUCs reported in Figure 4.1 suggests a potential relationship between firm identity information and predictive performance. The substantially higher firm classification performance of the "raw news" embeddings coincides with their higher AUC, whereas the lower classification performance of the "pure news" embeddings coincides with a lower AUC. This observation raises the possibility that the removal of firm-specific information may contribute to the difference in predictive performance between the two representations.

However, the classification results alone do not establish whether the amount of firm identity information encoded in an embedding is related to its predictive performance. To

investigate this relationship further, the same classification procedure is repeated separately for each news source type and news category. For a given source type or category, the embeddings are first restricted to the corresponding subset of news articles. For example, for the Valuation category, only articles tagged as Valuation are retained. The resulting observations are then divided according to the same chronological train-test split described in Section 4.6.2. The classifier is trained using the Valuation embeddings in the training window and subsequently evaluated on the Valuation embeddings in the test window, with the resulting balanced accuracy measuring how effectively firm identity can be inferred from this particular subset of news. The same procedure is repeated for both "raw news" and "pure news" embeddings across all news source types and news categories.

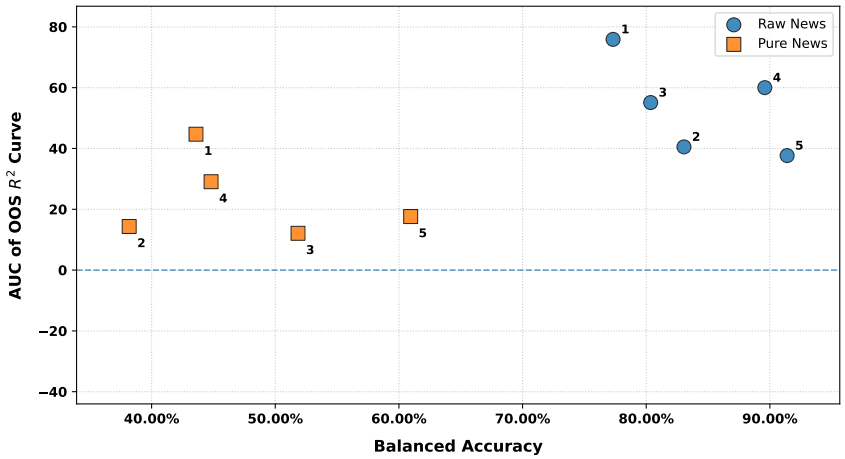
Rather than considering the resulting balanced accuracies in isolation, they are compared with the corresponding AUCs obtained from the predictive analysis. To examine the relationship between the two measures, we construct scatter plots with balanced accuracy on the horizontal axis and the corresponding AUC on the vertical axis for each representation and forecasting horizon. The resulting relationships for the different news source types and news categories are presented in Figure 4.2 and Figure 4.3, respectively.

For the news source types, across the 1-day, 5-day, and 21-day forecasting horizons, the "raw news" and "pure news" embeddings form visually distinct clusters. At the 1-day horizon, the "raw news" cluster is characterised by high balanced accuracy and high AUC, while the "pure news" cluster is characterised by low balanced accuracy and low AUC. At the 5-day horizon, the same clustering pattern is present. However, the "raw news" cluster shifts downward in AUC relative to the 1-day horizon, while the "pure news" cluster remains in approximately the same region of the plot. Since the balanced accuracy values for a given source type do not change across horizons, the horizontal position of both clusters remains broadly similar across panels, and the visible movement between horizons occurs only along the AUC axis. So, the downward shift of the "raw news" cluster at the 5-day horizon suggests that, at least visually, the AUC of the "raw news" embeddings varies more across horizons than that of the "pure news" embeddings. At the 21-day horizon, the same two-cluster structure is again observed, but the pattern is reversed relative to the 1-day and 5-day horizons. The "raw news" cluster is now positioned lower in AUC, while the "pure news" cluster is positioned higher.

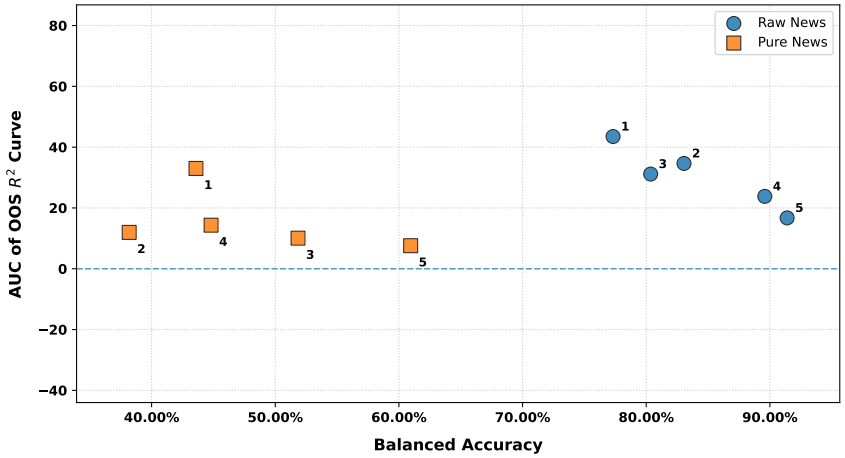
For the news tag categories, the clustering between "raw news" and "pure news" is visually less pronounced than for the news source types. At the 1-day and 5-day horizons, most "raw news" points are positioned at high balanced accuracy and high AUC, with Macro as a notable exception. At both horizons, similar to the relationship seen across the news source types, the "raw news" and "pure news" points appear to follow an approximately increasing relationship between balanced accuracy and AUC. This relationship appears more pronounced at the 1-day horizon, and the slope of this relationship appears to flatten at the 5-day horizon, largely driven by a decrease in the AUC of the "raw news" points, i.e., we again see the larger downward shift of the "raw news" AUC compared to the "pure news" AUC. At the 21-day horizon, no clear relationship between balanced accuracy and AUC is visually apparent, with the points appearing scattered across the plot.

To quantify the strength and statistical significance of the relationship, the balanced accuracy–AUC pairs obtained across the news source types and news categories are pooled, and the Pearson correlation coefficient is computed. The corresponding correlation coefficients and p-values are reported in Table 4.3.

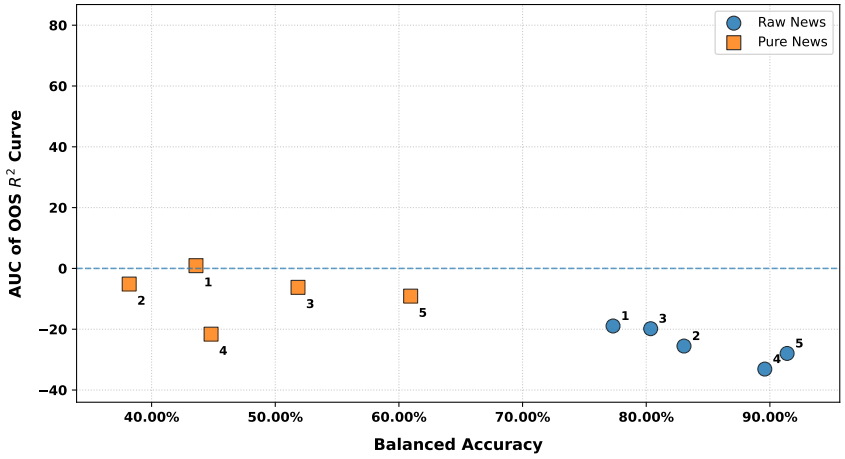
Using a significance level of $\alpha = 0.05$, the results in Table 4.3 show a positive Pearson correlation between balanced accuracy and AUC at the 1-day horizon ($r = 0.5515$, $p = 0.0011$), which is statistically significant. At the 5-day horizon, the correlation remains positive ($r = 0.4207$, $p = 0.0165$) and is also statistically significant, although lower in



(a) 1-day forecasting horizon

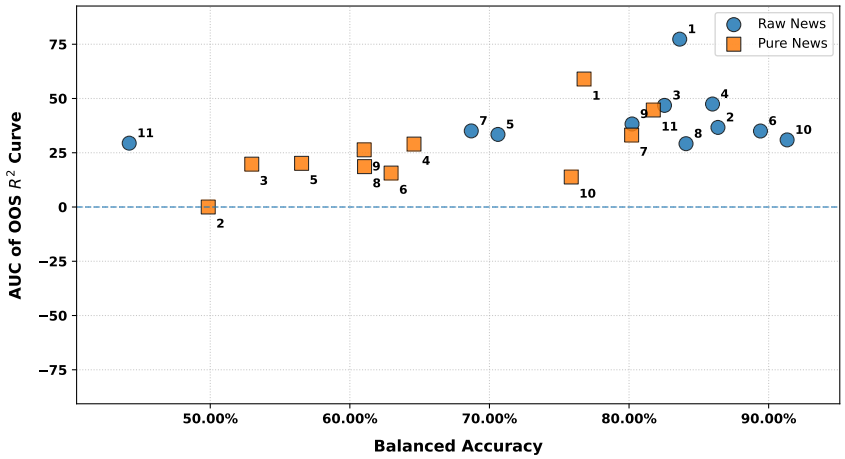


(b) 5-day forecasting horizon

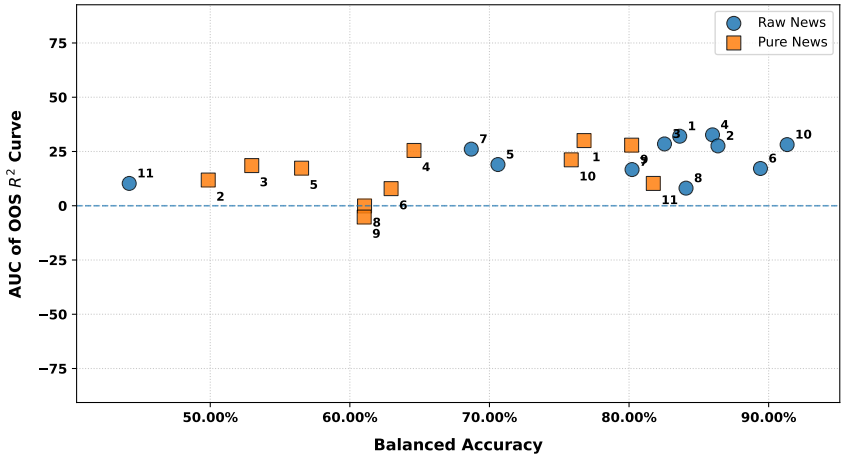


(c) 21-day forecasting horizon

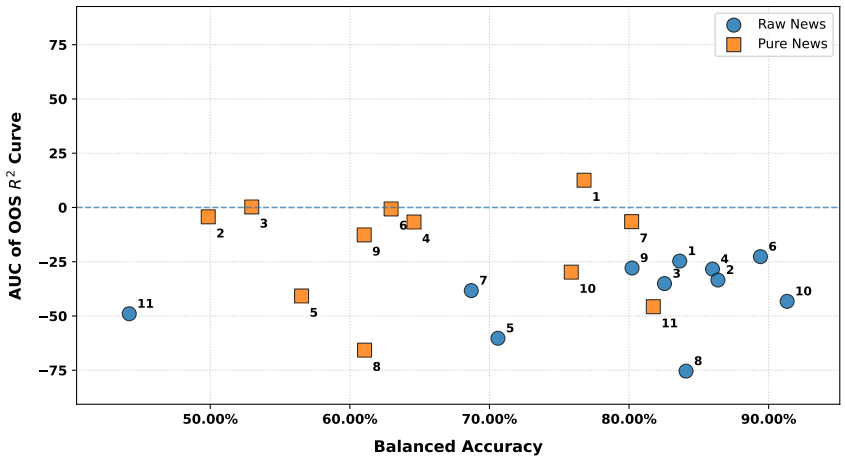
Figure 4.2: Relationship between balanced accuracy and AUC across news source types for the "raw news" and "pure news" embeddings. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively. Numerical labels denote: (1) Financial Media, (2) Mainstream, (3) Research, (4) Social Media, and (5) Trade Media.



(a) 1-day forecasting horizon



(b) 5-day forecasting horizon



(c) 21-day forecasting horizon

Figure 4.3: Relationship between balanced accuracy and AUC across news tags for the "raw news" and "pure news" embeddings. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively. Numerical labels denote: (1) Valuation, (2) Strategy, (3) Outlook, (4) Earnings, (5) Competition, (6) Operations, (7) Momentum, (8) Launches, (9) Legal Regulatory, (10) Analyst Ratings, and (11) Macro.

Table 4.3: Pearson correlation between balanced accuracy and AUC across news source types and news tags.

Horizon	Pearson Correlation	p-value
1-day	0.5515	0.0011
5-day	0.4207	0.0165
21-day	-0.3190	0.0752

magnitude than at the 1-day horizon. At the 21-day horizon, the correlation coefficient is negative ($r = -0.3190$, $p = 0.0752$) and is not statistically significant at the 5% level.

Hence, the classification results and the correlation analysis provide an empirical basis for the possibility that the amount of firm-specific information encoded in an embedding is associated with its predictive performance. This association appears strongest at the 1-day horizon, weaker but still statistically significant at the 5-day horizon, and is not present at the 21-day horizon. It is important, again, to emphasise that this analysis establishes an association rather than a causal relationship. The observed correlation between balanced accuracy and AUC may be driven by factors other than firm identity itself, and a detailed investigation of the underlying mechanism is beyond the scope of this study. Nonetheless, the consistency of a positive linear association at the shorter forecasting horizons offers empirical support for the view that firm-specific information encoded in the "raw news" embeddings may contribute, at least in part, to their higher predictive performance relative to the "pure news" embeddings.

These correlations may be confounded by differences in article volume across news source types and tags. Thus, a robustness check controlling for this factor is reported in Appendix D.1 and confirms that the association is robust at the 1-day horizon but weakens substantially at longer horizons.

4.6.5 Augmenting Pure News with Firm Identity

Throughout Section 4.6, we have established an association between the predictive performance of an embedding and the amount of information it contains about firm identity. Thus, the "pure news" embeddings, which are constructed using a residualisation procedure that substantially reduces the information about firm identity contained in them, achieve lower predictive performance than the "raw news" embeddings.

This raises a natural question: how would the predictive performance of "pure news" change if the model were explicitly provided with the identity of the firm? If the association established above reflects the role of firm identity in predictive performance, then providing the "pure news" representation with explicit information about firm identity should allow it to recover some, or potentially all, of the predictive performance observed for "raw news". We therefore test this possibility directly by augmenting the "pure news" regression with firm fixed effects.

To conduct this test, we repeat the PCA regression analysis described in Section 4.2, following the experimental design described in Section 4.3. The aggregation of the news embeddings, the PCA procedure, the training and test split, and the out-of-sample evaluation remain unchanged. For the "raw news" embeddings, we estimate the same regression specification as in Equation 4.2.4. For the "pure news" embeddings, however, we additionally include a firm fixed effect following the PCA transformation. The resulting specification is

$$y_{i,t}^{(h)} = \alpha + \gamma_i + \left(z_{i,t}^{P}(k)\right)^{\top} \beta^{P} + u_{i,t}^{(h)}, \tag{4.6.2}$$

where $y_{i,t}^{(h)}$ denotes the dependent variable for firm i at date t and forecasting horizon h, $z_{i,t}^{P}(k)$ denotes the first k principal components of the "pure news" embedding, β^P is the corresponding coefficient vector, and γ_i denotes the firm fixed effect.

The fixed effect γ_i assigns a separate coefficient to each firm. Let $\mathbf{1}_{\{i=c\}}$ denote an indicator function that takes the value 1 if observation i belongs to firm c and 0 otherwise. Then γ_i can be written as

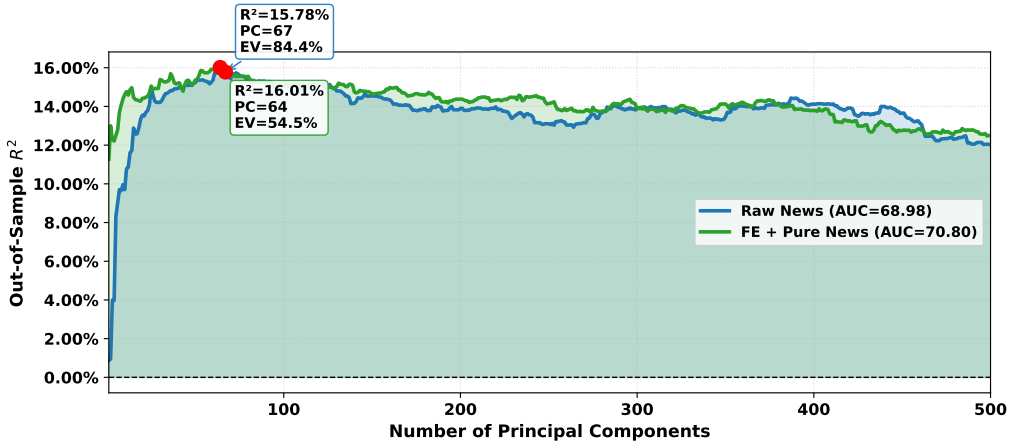
$$\gamma_i = \sum_{c=1}^{C} \gamma_c \, \mathbf{1}_{\{i=c\}},$$

so that each firm is assigned its own coefficient γ_c. These coefficients are estimated jointly with α and β^P by minimising the sum of squared residuals over the training sample, following the same procedure as Equation 4.2.5. The estimated coefficients $\hat{\gamma}_1, \ldots, \hat{\gamma}_C$ are then applied in Equation 4.6.2 to generate predictions for the test sample. This is equivalent to setting $\hat{\alpha} + \hat{\gamma}_c$ equal to firm c's mean value of $y_{i,t}^{(h)}$ in the training sample, so that the test-sample predictions use this training-sample firm mean as the firm-specific component. Thus, $\hat{\gamma}_i$ provides the regression with information about the identity of the firm associated with each observation.

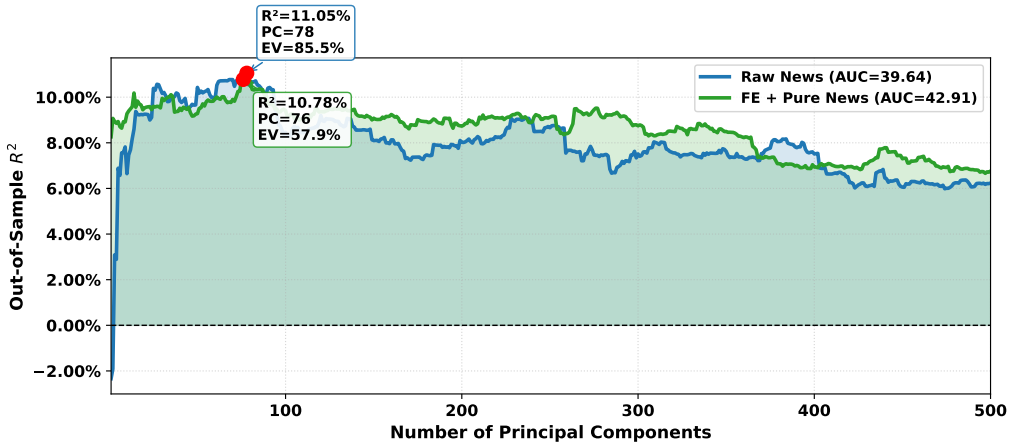
The "raw news" and "fixed effects + pure news" specifications are then evaluated using the criterion described in Section 4.3.5. The out-of-sample R^2 is calculated for each number of principal components, producing an R^2 profile as a function of the number of components, and the area under this profile (AUC) is then calculated, as illustrated previously in Figure 4.1. If the association established in Section 4.6 holds, the "fixed effects + pure news" specification should recover AUCs and R^2 profiles similar to those of "raw news", since the fixed effect restores the firm-identity information to the "pure news" embedding specification.

Figure 4.4 presents the AUC of the "raw news" and "fixed effects + pure news" regressions across the 1-day, 5-day, and 21-day forecasting horizons.

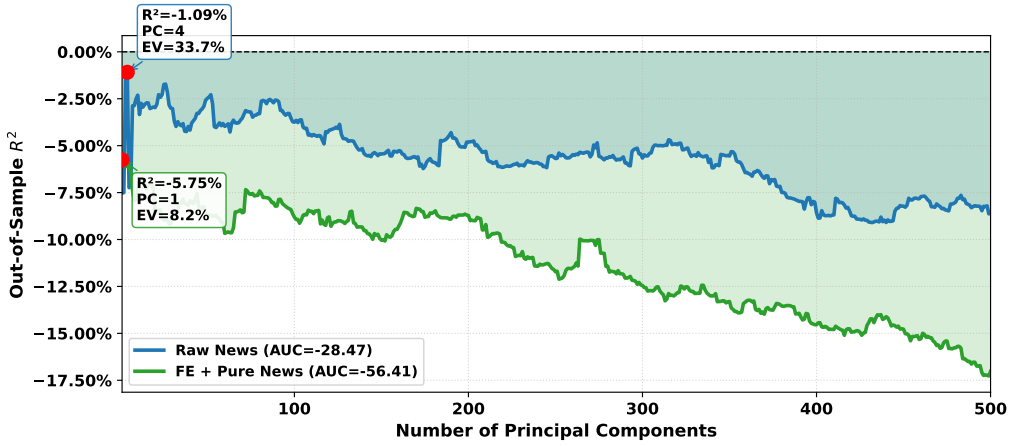
At the 1-day and 5-day forecasting horizons, the out-of-sample R^2 profiles of "raw news" and "fixed effects + pure news" are very similar, and the two specifications achieve close AUC values: 68.98 and 70.80, respectively, at the 1-day horizon, and 39.64 and 42.91, respectively, at the 5-day horizon. At both horizons, "fixed effects + pure news" performs marginally better than "raw news". It is also worth noting that, at both horizons, the best-performing number of principal components for the two specifications falls within a similar region, with the two specifications achieving similar maximum out-of-sample R^2 values: 15.78% (at 67 principal components) for "raw news" compared with 16.01% (at 64 principal components) for "fixed effects + pure news" at the 1-day horizon, and 11.05% (at 78 principal components) compared with 10.78% (at 76 principal components) at the 5-day horizon. A further notable observation is that "fixed effects + pure news" achieves this performance using a smaller proportion of explained variance than "raw news", suggesting that the firm fixed effect is doing much of the additional work that would otherwise require a larger number of principal components to capture. However, the pattern reverses at the 21-day forecasting horizon, where "fixed effects + pure news" performs considerably worse than "raw news", with both specifications achieving strongly negative AUC values. The AUC for "fixed effects + pure news" falls to -56.41, compared with -28.47 for "raw news". This represents a substantial deterioration relative to the original "pure news" specification without the fixed effect, whose AUC at this horizon was -11.13 (see Figure 4.1). This suggests that, at the 21-day horizon, the added firm identity information introduces a considerable amount of noise into the out-of-sample prediction rather than improving its performance.



(a) 1-day forecasting horizon



(b) 5-day forecasting horizon



(c) 21-day forecasting horizon

Figure 4.4: Out-of-sample R^2 profiles and AUC evaluation for "raw news" and "fixed effects + pure news" embeddings. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.

Chapter 5

Discussion

5.1 Further Elaboration on Results

One way of interpreting the prediction target used in this study is as a measure of the magnitude of a stock's abnormal price movement. Absolute abnormal return does not measure volatility in the strict statistical sense, but it provides a proxy for the size of the stock-specific price movement over a given forecasting horizon. Viewed in this way, the results suggest that, at shorter horizons, a substantial portion of the information useful for identifying the magnitude of a firm's abnormal price movement may be contained in information that is associated with the identity of the firm itself. This interpretation is consistent with our analysis using a multi-class classifier, in which "raw news" achieved substantially greater firm-identification accuracy than "pure news", as well as with the positive association between firm identifiability and predictive performance at the 1-day and 5-day horizons. Further support is provided when firm identity is explicitly reintroduced into the "pure news" representation, as the resulting model produces predictive performance that is very similar to that of "raw news" at the shorter horizons. Particularly, the AUC of the "fixed effects + pure news" specification is 70.80 at the 1-day horizon and 42.91 at the 5-day horizon, compared with 68.98 and 39.64, respectively, for "raw news". This suggests that the firm-specific component removed during residualisation contains information that is relevant to the prediction of the magnitude of short-term abnormal price movements. This interpretation should nevertheless be treated as an association rather than a causal conclusion, since firm identity may capture other persistent characteristics of firms that are themselves related to their responses to news.

This interpretation does not imply that firm-specific information contained in news is the only component relevant for return predictability. Such a conclusion would, in fact, be inconsistent with the findings of [4]. Their analysis began from the observation that news embeddings contain a component that is predictable from prevailing stock characteristics and separates this component from the residual "pure news" component. They subsequently examined the ability of these representations to predict subsequent signed excess returns using a monthly stock-level framework. In contrast, our study asks a different question: rather than examining whether news can be used to predict the direction and persistence of subsequent returns, it examines the extent to which news embeddings can identify the magnitude of a stock-specific abnormal price movement over short daily horizons. This distinction is important when interpreting the apparently different findings. [4] found that the residual "pure news" component contains substantial return-predictive information and that its predictive content remains statistically significant for up to approximately 18 months. Their results therefore indicated that the information left after removing predictable firm-characteristic content is not simply irrelevant information. Rather, it represents information that may be incorporated into prices gradually. Our

findings can be viewed as complementary to this result. The stronger performance of "raw news" at the shorter horizons suggests that information associated with firm identity is particularly useful for identifying the immediate magnitude of a stock's abnormal response, while the disappearance of the advantage of "raw news" at the 21-day horizon is consistent with this firm-specific component becoming less informative over longer horizons. Importantly, however, our study does not establish that "pure news" ceases to contain predictive information after 21 days. Since our study does not examine signed returns or horizons extending beyond 21 trading days, it cannot directly test the longer-lived return predictability documented by [4]. Instead, the two studies may be capturing different dimensions of the information contained in news, i.e., firm-specific information may be particularly useful for identifying the magnitude of an immediate price response, while residual news may contain information relevant to the subsequent direction and persistence of returns.

The predictive relationship also varies according to the source and economic content of the news, suggesting that the role of firm-specific information is unlikely to be uniform across all forms of news. Financial Media provides the clearest example. It produces the strongest predictive performance among the five source types at both the 1-day and 5-day horizons, with "raw news" achieving AUC values of 75.89 and 43.49, respectively. One possible explanation is that financial media is more closely connected to firm-specific developments and therefore contains information that allows the model to identify not only the occurrence of news but also the particular firm whose characteristics determine the magnitude of its response. In such circumstances, retaining firm identity may provide substantial information about the expected size of the subsequent abnormal movement. The tag-level results similarly demonstrate that the effect is not universal. Valuation news produces the strongest short-horizon performance for both representations, whereas Macro news provides an important exception at the 1-day horizon, where "pure news" outperforms "raw news". At the 21-day horizon, "pure news" performs better than "raw news" across all eleven news categories, although predictive performance is generally weak. These differences suggest that the predictive relevance of firm-specific information depends partly on the economic nature of the information being conveyed. This is broadly consistent with [4], who find that the market response to news differs substantially across economic themes and identify distinct patterns of underreaction and overreaction across different types of news. Their analysis therefore reinforces the broader implication that news should not be treated as a homogeneous information source, which is consistent with our results. Different components of news may contain information that becomes relevant at different stages of the price-adjustment process, and the relative importance of firm-specific and residual information may consequently vary both across news types and across forecasting horizons.

5.2 Limitations and Future Work

The findings of this study should be interpreted in light of several limitations arising from the available data and the empirical design. These limitations do not invalidate the results, but they constrain the extent to which the findings can be generalised and the mechanisms underlying the observed relationships can be established. At the same time, each limitation identifies a direction in which the analysis could be extended.

The most immediate limitation concerns the time period considered in the final analysis. The final predictive sample covers only the period from February 2024 to July 2025. This relatively short period necessarily limits the number of stock-price observations available for estimating the relationship between news embeddings and subsequent abnormal returns. The limited number of observations may reduce the precision of the estimated

predictive relationships and increase the sensitivity of the results to individual observations. More importantly, the period considered may represent only a particular market regime. As a result, it is possible that some of the patterns identified in this study are specific to the market environment prevailing during the sample period rather than being stable characteristics of the relationship between news and abnormal returns. Extending the analysis to a substantially longer historical period would therefore provide a larger number of observations and allow the stability of the results to be examined across different market regimes. Therefore, repeating the analysis over periods characterised by substantially different levels of market volatility would help determine whether the observed relationship between "raw news", "pure news", and absolute abnormal returns is persistent or regime-dependent.

A further limitation arises from the temporal resolution of the news data. The initial objective of the study was to investigate whether the information contained in news could be used to estimate a price-impact kernel, thereby examining how the information conveyed by an article translates into stock-price movements over time. However, the available news data did not provide sufficiently precise timestamps for the individual articles to permit such an analysis at the article level. Hence, articles relating to a particular stock were aggregated to a common daily observation. While this aggregation allowed the news data to be aligned with daily stock-price observations, it may also have obscured variation in the timing and sequence of information arrival within the trading day. Different articles published on the same day may have generated substantially different price responses depending on when the information became available to the market, and aggregating them into a single daily embedding may therefore have discarded some of the temporal structure contained in the original news. The absence of sufficiently granular timestamps also constrained the estimation of a price-impact kernel. Given the relatively small number of observations available, estimating the response of prices over many post-news time intervals would require the estimation of a large number of parameters relative to the available data, creating a substantial risk of overfitting. For this reason, the analysis was restricted to daily predictive horizons rather than attempting to estimate the full price-impact kernel. A natural extension would therefore be to obtain news data with precise article-level timestamps and combine it with a substantially larger sample of firms and observations. This would permit the estimation of a more granular price-impact kernel and allow the speed, persistence, and shape of the market's response to different components of news to be examined directly.

The limited cross-sectional dimension of the study is another important constraint. The final analysis considers only 14 stocks. While this restriction was necessary given the availability and quality of the data, it limits the generalisability of the findings and reduces the ability to distinguish firm-specific effects from broader patterns. A larger cross-section would provide substantially more observations and would make it possible to determine whether the relationship identified in this study persists across a wider range of firms, industries, and market characteristics. The limited cross-section also constrained the analysis of the information contained in firm identity. The present analysis examines whether the amount of firm-specific information retained in an embedding is associated with predictive performance, but it does not separately investigate whether industry-level information or other forms of group-level information contribute to the predictive relationship. For example, news concerning firms in the technology sector may contain common informational features that are not captured simply by identifying the individual firm. A substantially larger universe would therefore allow the analysis to distinguish between information associated with individual firms, industries, and broader economic sectors. It would also provide a stronger basis for evaluating whether the observed differences between "raw news" and "pure news" are robust across different types of firms rather than

being driven by the particular firms included in the present sample.

Another limitation is that our analysis establishes that "raw news" and "pure news" differ in their ability to predict absolute abnormal returns, but does not directly identify what information within the embeddings generates this predictive variation. Therefore, while the results demonstrate that the representations contain predictive information, they do not establish whether this information relates primarily to sentiment, valuation, financial performance, management actions, industry conditions, uncertainty, or other textual characteristics. This limits the ability to provide an economic interpretation of the mechanism through which the embeddings generate predictive performance. Future research could therefore employ interpretability techniques, including sparse autoencoders (SAEs) or other methods for decomposing embedding representations into more interpretable features. Such approaches could be used to investigate which latent features are activated by "raw news" and "pure news" and whether particular features are systematically associated with larger subsequent abnormal price movements.

Another concern is the interpretation of the differences in predictive performance across forecasting horizons. While the empirical results clearly indicate that "raw news" performs better at the 1-day horizon than at the 5-day horizon, and that predictive performance deteriorates substantially by the 21-day horizon, our analysis does not establish the precise mechanism responsible for these differences. Specifically, it is not possible to determine from the current results whether the stronger performance at shorter horizons reflects the speed with which information is incorporated into prices, the greater relevance of firm-specific information to immediate price movements, or other characteristics of the news and market environment. The relatively weak performance at the 21-day horizon warrants particular caution in interpretation. The out-of-sample period extends from 1 April 2025 to 23 July 2025, and constructing a 21-day-ahead return necessarily reduces the number of observations that can be used to evaluate this horizon relative to shorter horizons. The resulting reduction in the effective number of observations may increase the noise in the estimated predictive relationship and may therefore contribute to the weaker performance observed at the 21-day horizon. Thus, the deterioration in predictive performance at longer horizons should not be interpreted solely as evidence that the predictive information contained in the embeddings disappears over time. A longer sample with substantially more observations would allow the differences between the 1-day, 5-day, and 21-day horizons to be examined with greater statistical precision. Future research could further extend the analysis to a wider range of horizons and investigate whether the decline in predictive performance occurs gradually or reflects a more specific point at which the predictive content of different components of news changes.

A further potential limitation concerns the embedding model used in the analysis. The study relies on the `text-embedding-3-large` model to represent the news articles and it is possible that the parameters of this pretrained embedding model may have been estimated using text that extends beyond the period used in our study. As a result, although our study has attempted to minimise look-ahead bias through chronological sample splitting, the embedding model may potentially contain information derived from text published during, or after, the period treated as the out-of-sample test period. This creates a potential source of look-ahead information that is external to the predictive regression itself. Future research could address this issue by using an embedding model whose parameters are trained exclusively on text available prior to the beginning of the relevant out-of-sample period, or by using chronologically consistent embedding models whose training process explicitly prevents exposure to future information. Repeating the analysis with such models would provide a stronger test of whether the observed predictive relationship survives when potential look-ahead information within the representation itself is eliminated.

Our results also suggest a potential application beyond the prediction of absolute ab-

normal returns. Since the dependent variable measures the magnitude of abnormal price movements, the signal identified in this study may have relevance for financial instruments whose value depends on the occurrence or magnitude of large price movements rather than solely on their direction. One possible extension would therefore be to investigate whether news embeddings can be used to identify situations in which the expected magnitude of a stock-price movement is sufficiently large to make volatility-sensitive strategies economically attractive. For example, a long straddle involves simultaneously purchasing a call and a put with the same underlying asset and expiration, and therefore benefits from a sufficiently large movement in the underlying price in either direction. If news embeddings can reliably identify stocks that are likely to experience unusually large abnormal movements, this information could potentially be examined as a signal for identifying situations in which the expected realised movement of the underlying may exceed the movement implied by option prices. However, such an application would require substantial additional analysis since absolute abnormal return is not equivalent to implied or realised volatility and option profitability also depends on factors such as the option premium, strike price, time to maturity and other factors. Future research could therefore investigate whether the predictive signal identified in this study contains incremental information for option-implied volatility or realised volatility and whether it can be translated into economically meaningful trading strategies such as volatility or straddle-based strategies.

Chapter 6

Conclusion

This thesis aimed to compare the predictive performance of "raw news" and "pure news" embeddings in predicting the magnitude of subsequent abnormal price movements, and to examine the role of firm identity in their predictive performance.

Our study first established the extent to which variation in news embeddings can be associated with observable firm characteristics. Following the preprocessing of the raw news embeddings, a regularised regression framework was used to remove the component of the embeddings that could be explained by the JKP firm characteristics. The resulting residual representation was subsequently compared with a within-firm specification, and the results indicated that the variation removed under the two approaches was closely related to persistent firm identity. On this basis, firm fixed-effects residual embeddings were used as the primary representation of "pure news" for the subsequent predictive analysis.

The comparison of "raw news" and "pure news" showed that "raw news" generally provides stronger predictive performance at the 1-day and 5-day horizons, while both representations perform substantially worse at the 21-day horizon, where "pure news" generally performs somewhat better. This pattern is broadly consistent across news source types and news categories, although some differences emerge across individual categories. Further analysis showed that the greater predictive performance of "raw news" at shorter horizons is associated with the amount of firm-specific information retained in the embeddings. Finally, reintroducing firm identity into the "pure news" representation largely restored the predictive performance of "raw news" at the 1-day and 5-day horizons.

The findings should nevertheless be interpreted in light of the limited cross-section and observation period, as well as the absence of a direct identification of the economic information encoded within the embeddings. Future research using a larger universe of firms, more granular news timestamps, and more interpretable representations could provide further insight into how different components of financial news are incorporated into asset prices and how their predictive relevance evolves over time.

Appendix A

Transformer Architecture

Earlier neural network language models, such as recurrent neural networks (RNNs), processed token sequences sequentially, where each token representation was generated based on information from previously processed tokens. In this context, a sequence refers to an ordered collection of tokens. Although they were effective in certain applications, these models faced limitations in capturing long-range dependencies, as information from earlier tokens had to be propagated through the sequence step by step. Moreover, contextual embeddings require a model that can incorporate information from surrounding tokens when generating a representation of each token. The Transformer architecture, introduced by [27], addressed these limitations by replacing recurrence with attention mechanisms. This allows each token to directly incorporate information from other tokens in the sequence, enabling the construction of contextual representations.

Input Embeddings and Positional Encoding

Before being processed by the Transformer layers, tokens are mapped to token embeddings, which provide the initial numerical representation of the input sequence.

Since Transformers process all tokens in a sequence simultaneously, they do not naturally contain information about the order of tokens. To incorporate positional information, positional encodings are added to the token embeddings. The resulting input representation is given by:

$$X = E + P$$

where E represents the matrix of token embeddings and P represents the corresponding positional encodings. Therefore, the input to the Transformer contains both semantic information from the token embeddings and information about the position of each token in the sequence [27].

In the original Transformer architecture, positional encodings were generated using sinusoidal functions of different frequencies [27]. Later Transformer-based models have also employed learned positional embeddings, where positional information is learned as model parameters during training [3].

Encoder and Decoder

The original Transformer architecture consists of two main components: an encoder and a decoder. The encoder processes an input sequence and produces a set of contextual representations, whereas the decoder generates an output sequence based on these representations [27].

The encoder is composed of a stack of identical layers. Each layer contains two main sub-layers where the first is a multi-head self-attention mechanism, which is the key innovation of the Transformer architecture. This mechanism allows each token representation to incorporate information from other tokens in the sequence. The second sub-layer is a position-wise feed-forward neural network, which applies further nonlinear transformations to the token representations. Additionally, the input to each sub-layer is added to its output. These connections, known as residual connections, help preserve information from earlier layers and improve the flow of information through the network. Following each residual connection, to stabilise the distribution of the resulting representations before they are passed to subsequent layers, layer normalisation is applied [27].

The decoder is also composed of a stack of identical layers. However, unlike the encoder, each decoder layer contains an additional encoder-decoder attention sub-layer, which allows the decoder to incorporate information from the encoder output when generating the output sequence. Furthermore, the decoder applies masked self-attention, which prevents a token from attending to future tokens in the sequence. This restriction ensures that the model generates outputs autoregressively, where each prediction depends only on previously generated tokens [27].

Although the original Transformer architecture contains both an encoder and a decoder, modern language models commonly adopt only one component depending on their intended application. Encoder-only models, such as BERT, are designed to learn contextual representations of input text, while decoder-only models, such as GPT, are primarily designed for autoregressive text generation. For representation learning tasks, encoder-based architectures are used to obtain vector representations that capture the semantic information contained within text [3, 20].

Self-Attention Mechanism

The self-attention mechanism is the fundamental operation that enables Transformers to construct contextual representations. Unlike recurrent models, where information is propagated sequentially, self-attention allows every token to directly interact with every other token in the sequence [27].

Let X denote the matrix of input token representations, where each row corresponds to the representation of an individual token. The self-attention mechanism first projects X into three separate matrices known as queries, keys, and values:

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V,$$

where W_Q, W_K, and W_V are learned parameter matrices. The query and key vectors determine the relevance between pairs of tokens, while the value vectors contain the information that is aggregated according to these relationships. The attention output is computed using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V.$$

The matrix product QK^T calculates similarity scores between each query and all keys. These scores are scaled by $\sqrt{d_k}$ to prevent excessively large values from causing unstable gradients during optimisation. The softmax operation converts the similarity scores into attention weights, which determine the contribution of each value vector to the updated representation. Therefore, the output representation of each token contains information from other relevant tokens in the sequence [27].

The Transformer extends scaled dot-product attention through multi-head attention, where multiple attention operations are performed using different learned projections of

queries, keys, and values. Each attention head can learn different types of relationships between tokens, allowing the model to capture multiple aspects of the input sequence simultaneously. The outputs of the individual attention heads are concatenated and linearly transformed to produce the final representation [27].

Output Representations

The output produced by a Transformer depends on the objective for which the model is trained. In generative language models, such as GPT, the final contextual representations produced by the Transformer are passed through a linear output layer that computes a score for every token in the model's vocabulary. These scores, known as logits, are unnormalised values that indicate how likely each vocabulary token is to be the next token in the sequence. A softmax function then converts these logits into probabilities. This allows the model to predict the next token [20]. However, not all Transformer-based models are designed for text generation. Encoder-based models produce contextual representations for each input token rather than predicting the next token [3]. These representations can be used directly for downstream tasks or combined to produce a fixed-dimensional embedding representing the entire input text [21]. Such embeddings can then be used for tasks such as semantic similarity, retrieval, and classification.

Appendix B

Additional Tables

B.1 List of Selected Large-Cap Stocks

Table B.1: Selected Large-Cap Stocks

Company	**Ticker**	**Industry**	**Market Cap (USD)**
Nvidia	NVDA	Semiconductors	$4.4 trillion
Apple	AAPL	Technology (Consumer Electronics)	$3.8 trillion
Alphabet	GOOG	Media / Internet Services	$3.7 trillion
Microsoft	MSFT	Software	$3.0 trillion
Amazon	AMZN	Retail (E-commerce)	$2.1 trillion
Meta Platforms	META	Media / Internet Services	$1.6 trillion
Broadcom	AVGO	Semiconductors	$1.6 trillion
Tesla	TSLA	Automobiles	$1.6 trillion
Berkshire Hathaway	BRK-B	Diversified Financials	$1.1 trillion
Walmart	WMT	Consumer Retailing	$1.1 trillion
Eli Lilly	LLY	Pharmaceuticals & Biotech	$927.5 billion
JPMorgan Chase	JPM	Banks	$816.0 billion
Visa	V	Diversified Financials	$617.9 billion
Oracle	ORCL	Software	$449.7 billion

Source: [23]. Berkshire Hathaway's figure reflects the company's total market capitalisation (BRK-B is the Class B share class of the same company). Market capitalisations fluctuate continuously with share price and should be treated as approximate at the time of retrieval.

B.2 List of News Tags

Table B.2: News Tag Classification

News Tag	**Meaning**
AI	News concerning artificial intelligence applications, developments, or products.
Accessibility	News concerning access to products, services, technologies, or markets.
Accounting	News concerning accounting practices, standards, reporting, or financial statements.

Continued on next page

Table B.2 – continued from previous page

News Tag	Meaning
Advertising	News concerning advertising, marketing campaigns, or promotional activities.
Analyst Ratings	Analyst recommendations, ratings, price targets, or changes in analyst views.
Artificial Intelligence	News concerning the development, deployment, or commercial use of artificial intelligence technologies.
Branding	News concerning brand development, reputation, positioning, or corporate identity.
Capital Returns	News concerning dividends, share repurchases, or other distributions of capital to shareholders.
Catalysts	Events or developments identified as potential drivers of future company or stock performance.
Compensation	News concerning executive or employee compensation, remuneration, or incentive schemes.
Competition	News concerning competitors, competitive positioning, or changes in the competitive environment.
Cybersecurity	News concerning cybersecurity incidents, threats, vulnerabilities, or security measures.
Deals	News concerning significant corporate or commercial transactions.
Digital Transformation	News concerning the adoption of digital technologies to transform business operations or processes.
Drug Pricing	News concerning pharmaceutical pricing, drug costs, or changes in pricing policy.
ESG	News concerning environmental, social, and governance issues.
Earnings	News concerning company earnings, profits, revenue, or quarterly and annual financial results.
Education	News concerning education, training, or developments in the education sector.
Equity	News concerning equity markets, shares, equity issuance, or ownership interests.
Financials	News concerning financial performance, balance sheets, cash flows, or other financial information.
Fraud	News concerning allegations, investigations, or incidents of fraudulent activity.
Governance	News concerning corporate governance, boards, management structures, or shareholder governance matters.
IP	News concerning intellectual property, patents, trademarks, copyrights, or related disputes.
Innovation and Technology	News concerning technological innovation, new technologies, or technology-driven developments.
Insider Activity	News concerning transactions or actions undertaken by company insiders.
Insider Buying	News concerning purchases of company securities by corporate insiders.
Insider Buying/Selling	News concerning both purchases and sales of company securities by insiders.
Insider Selling	News concerning sales of company securities by corporate insiders.
Insider Trading	News concerning alleged or confirmed trading based on material nonpublic information.
Insider Transactions	News concerning transactions in company securities conducted by insiders.
Institutional Holdings	News concerning ownership or holdings of shares by institutional investors.
Insurance	News concerning insurance products, companies, premiums, claims, or developments in the insurance industry.

Continued on next page

Table B.2 – continued from previous page

News Tag	Meaning
Investors	News concerning investors, investor activity, or investor relations.
Launches	News concerning the launch or introduction of new products, services, or initiatives.
Leadership	News concerning executives, senior management, appointments, departures, or leadership changes.
Legal	News concerning lawsuits, litigation, legal disputes, or other legal matters.
Regulatory	News concerning regulations, regulatory actions, approvals, or compliance requirements.
M&A	News concerning mergers, acquisitions, takeovers, or related corporate transactions.
Macro	News concerning broader macroeconomic conditions, including economic growth, inflation, interest rates, or employment.
Momentum	News concerning market or stock price momentum and related developments.
Operations	News concerning business operations, production, supply chains, or operational performance.
Outlook	News concerning management forecasts, expectations, guidance, or future business prospects.
Ownership	News concerning company ownership structures or changes in ownership.
Partnerships	News concerning strategic partnerships, alliances, or collaborations.
Policy	News concerning government policy, legislation, or policy changes affecting companies or markets.
Portfolio	News concerning investment portfolios, asset allocation, or portfolio holdings.
Prior Performance	News concerning historical company or asset performance.
Privacy	News concerning data privacy, personal information, or privacy regulations.
Product Development	News concerning the development, improvement, or testing of products and services.
Retail	News concerning retail businesses, consumers, or retail-sector activity.
Sentiment	News concerning market, investor, or consumer sentiment.
Risk	News concerning financial, operational, market, or other forms of risk.
Risk Management	News concerning measures, strategies, or systems used to identify and manage risk.
Strategy	News concerning corporate strategy, strategic priorities, or long-term business plans.
Tax	News concerning taxation, tax policy, tax liabilities, or changes in tax treatment.
TechnicalAnalysis	News concerning technical indicators, chart patterns, or technical analysis of financial markets.
Technology	News concerning technology products, services, infrastructure, or technological developments.
Valuation	News concerning company or asset valuation, valuation metrics, or changes in perceived value.
Wealth	News concerning wealth management, private wealth, or developments affecting high-net-worth investors.

B.3 List of News Source Types

Table B.3: News Source Type Classification

News Source Type	Meaning
Company IR	Information released directly by companies, such as earnings releases, filings, and corporate announcements.
Financial Media	Financial media covering markets, companies, investments, and financial events.
Research	Research reports, analytical publications, and data-oriented sources.
Trade Media	Industry-specific publications covering developments within particular sectors or professions.
Social Media	Social media content, blogs, and other informal or user-generated sources.
Regulators	Publications and announcements from financial or industry regulatory authorities.
Mainstream	General news outlets covering broad economic, political, and business events.
Newswires	News distributed through professional newswire services.
PR Wires	Press releases and corporate communications distributed through public-relations wire services.
Government Stats	Official statistics, economic data, and other information published by government agencies.

B.4 List of Selected JKP Characteristics

Table B.4: Selected JKP Characteristics and Descriptions

Variable	Description
prc	Price per share
market_equity	Market equity
div12m_me	Dividend yield over the previous 12 months
chcsho_12m	Change in shares over 12 months
eqnpo_12m	Net equity payout over 12 months
ret_1_0	Short-term reversal
ret_3_1	Momentum, months 1–3
ret_6_1	Momentum, months 1–6
ret_9_1	Momentum, months 1–9
ret_12_1	Momentum, months 1–12
ret_12_7	Price momentum, months 7–12
ret_60_12	Long-term reversal, months 12–60
seas_1_1an	One-year annual seasonality
seas_1_1na	One-year non-annual seasonality
seas_2_5an	Two-to-five-year annual seasonality
seas_2_5na	Two-to-five-year non-annual seasonality
at_gr1	Asset growth, one year
sale_gr1	Sales growth, one year
debt_gr3	Total debt growth, three years
sale_gr3	Sales growth, three years
fnl_gr1a	Financial liabilities change, one year, scaled by assets
nfna_gr1a	Net financial assets change, one year, scaled by assets

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Table B.4 – continued from previous page

Variable	Description
tax_gr1a	Effective tax rate change, one year
be_gr1a	Book equity change, one year, scaled by assets
ebit_sale	Operating profit margin after depreciation
gp_at	Gross profits-to-assets
cop_at	Cash-based operating profitability-to-assets
ni_be	Net income-to-book equity
ebit_bev	EBIT-to-book enterprise value
netis_at	Net issuance scaled by assets
eqnetis_at	Equity net issuance scaled by assets
dbnetis_at	Net debt issuance scaled by assets
oaccruals_at	Operating accruals-to-assets
oaccruals_ni	Percent operating accruals
taccruals_at	Total accruals-to-assets
taccruals_ni	Percent total accruals
opex_at	Operating leverage
at_turnover	Asset turnover
sale_bev	Sales scaled by book enterprise value
cash_at	Cash-to-assets
sale_emp_gr1	Sales growth per employee, one year
ni_inc8q	Number of consecutive earnings increases
lnoa_gr1a	Change in long-term net operating assets, one year, scaled by average assets
saleq_gr1	Quarterly sales growth, one year
niq_be	Quarterly return on equity
niq_at	Quarterly return on assets
niq_at_chg1	Change in quarterly return on assets
dsale_drec	Change in sales minus change in receivables
dgp_dsale	Change in gross margin minus change in sales
saleq_su	Standardized revenue surprise
niq_su	Standardized earnings surprise
op_atl1	Operating profits-to-lagged book assets
gp_atl1	Gross profits-to-lagged assets
cop_atl1	Cash-based operating profitability-to-lagged assets
pi_nix	Taxable income-to-book income
ocf_at	Operating cash flow-to-assets
op_at	Operating profits-to-assets
ocf_at_chg1	Change in operating cash flow-to-assets
at_be	Book leverage
ocfq_saleq_std	Quarterly operating cash-flow-to-sales volatility
earnings_variability	Earnings variability
kz_index	Kaplan–Zingales index
ni_ar1	One-year lagged net income-to-assets
ni_ivol	Net income idiosyncratic volatility
at_me	Assets-to-market equity
be_me	Book-to-market equity
debt_me	Debt-to-market equity
netdebt_me	Net debt-to-market equity
sale_me	Sales-to-market equity
ni_me	Earnings-to-price
ocf_me	Operating cash flow-to-market equity
fcf_me	Free cash flow-to-market equity
eqpo_me	Payout yield
eqnpo_me	Net payout yield
ival_me	Intrinsic value-to-market equity

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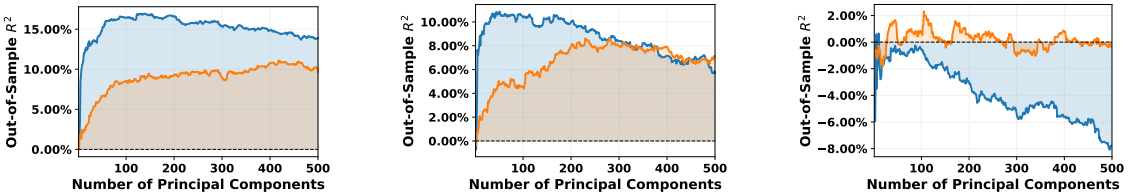
Table B.4 – continued from previous page

Variable	**Description**
bev_mev	Book enterprise value-to-market enterprise value
ebitda_mev	EBITDA-to-market enterprise value
eq_dur	Equity duration
beta_60m	60-month market beta
resff3_12_1	Residual momentum, 12 months to 1 month
resff3_6_1	Residual momentum, 6 months to 1 month
mispricing_mgmt	Management-based mispricing
mispricing_perf	Performance-based mispricing
ivol_capm_21d	CAPM idiosyncratic volatility, 21 days
iskew_capm_21d	CAPM idiosyncratic skewness, 21 days
coskew_21d	Coskewness, 21 days
beta_dimson_21d	Dimson beta, 21 days
ivol_ff3_21d	Fama–French three-factor idiosyncratic volatility, 21 days
iskew_ff3_21d	Fama–French three-factor idiosyncratic skewness, 21 days
ivol_hxz4_21d	Hou–Xue–Zhang q-factor idiosyncratic volatility, 21 days
iskew_hxz4_21d	Hou–Xue–Zhang q-factor idiosyncratic skewness, 21 days
rmax5_21d	Highest five daily returns, 21 days
rmax1_21d	Maximum daily return, 21 days
rvol_21d	Return volatility, 21 days
rskew_21d	Total return skewness, 21 days
zero_trades_21d	Number of zero-trading days, 21 days
dolvol_126d	Dollar trading volume, 126 days
dolvol_var_126d	Dollar trading-volume volatility, 126 days
turnover_126d	Share turnover, 126 days
turnover_var_126d	Share-turnover volatility, 126 days
zero_trades_126d	Number of zero-trading days, 126 days
zero_trades_252d	Number of zero-trading days, 252 days
ami_126d	Amihud illiquidity measure, 126 days
ivol_capm_252d	CAPM idiosyncratic volatility, 252 days
prc_highprc_252d	Current price relative to the previous-year high
betadown_252d	Downside beta, 252 days
bidaskhl_21d	High–low bid–ask spread, 21 days
corr_1260d	Correlation with the market, 1260 days
betabab_1260d	Frazzini–Pedersen betting-against-beta measure, 1260 days
rmax5_rvol_21d	Highest five daily returns scaled by volatility
age	Firm age
qmj	Quality Minus Junk, composite
qmj_prof	Quality Minus Junk, profitability
qmj_growth	Quality Minus Junk, growth
qmj_safety	Quality Minus Junk, safety
niq_be_chg1	Change in quarterly return on equity
seas_6_10an	Six-to-ten-year annual seasonality
seas_6_10na	Six-to-ten-year non-annual seasonality
lti_gr1a	Change in long-term investments, one year, scaled by assets

Appendix C

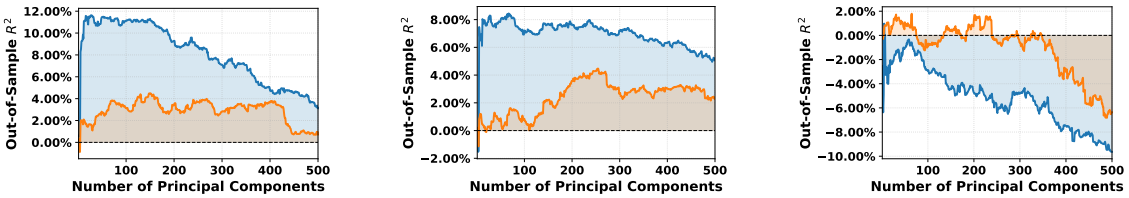
Additional Figures

C.1 Out-of-sample R^2 Profiles Across News Source Types



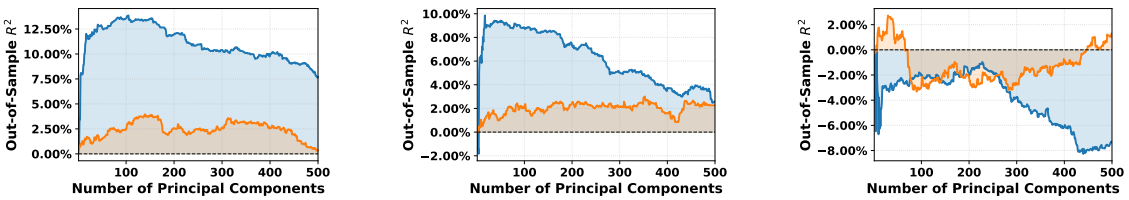
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.1: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings from Financial Media. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



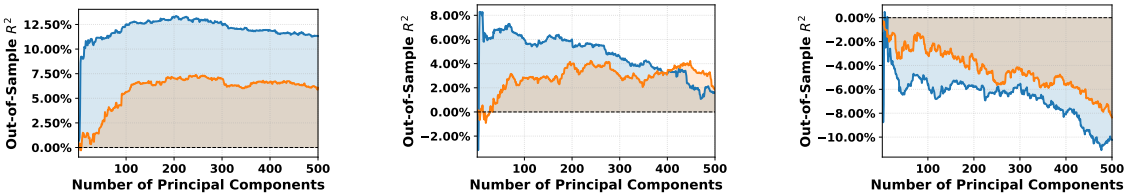
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.2: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings from Mainstream. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



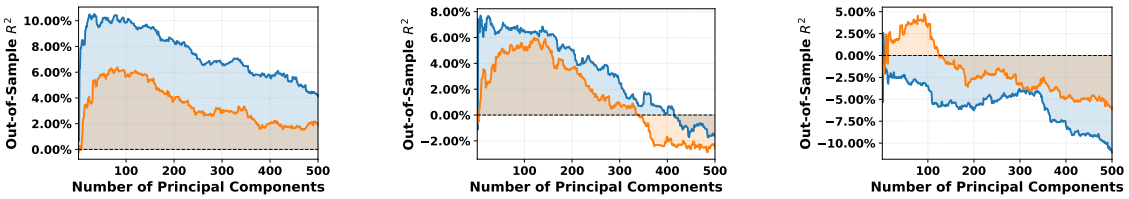
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.3: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings from Research. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

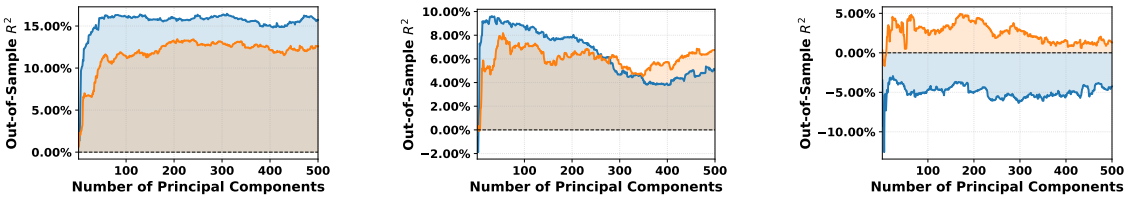
Figure C.4: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings from Social Media. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

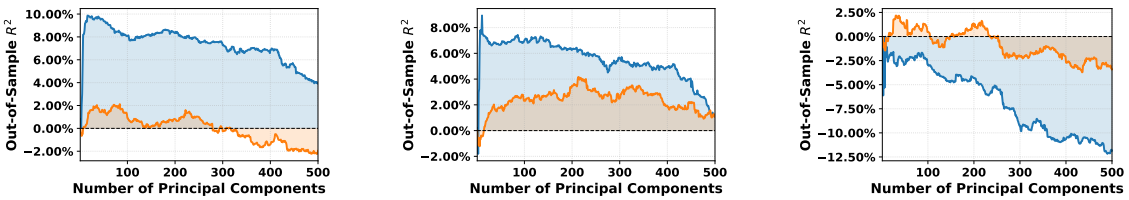
Figure C.5: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings from Trade Media. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.

C.2 Out-of-sample R^2 Profiles Across News Tags



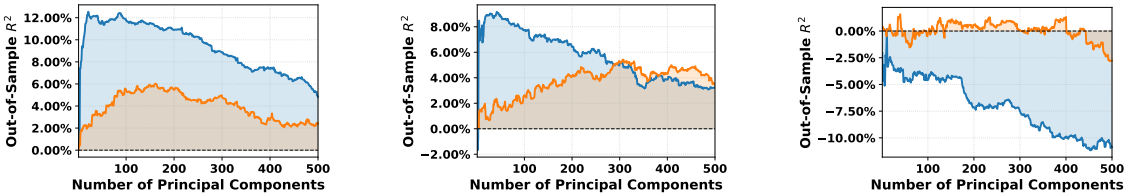
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.6: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Valuation news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



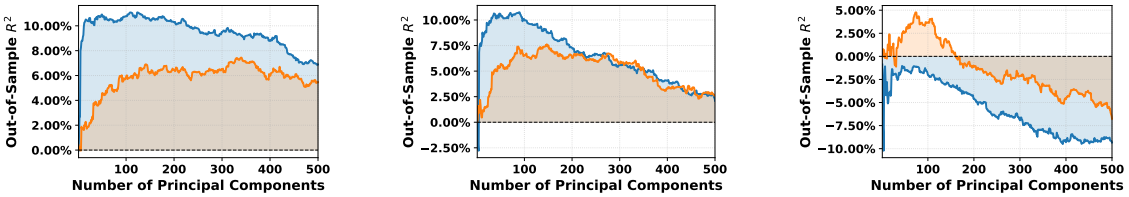
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.7: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Strategy news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



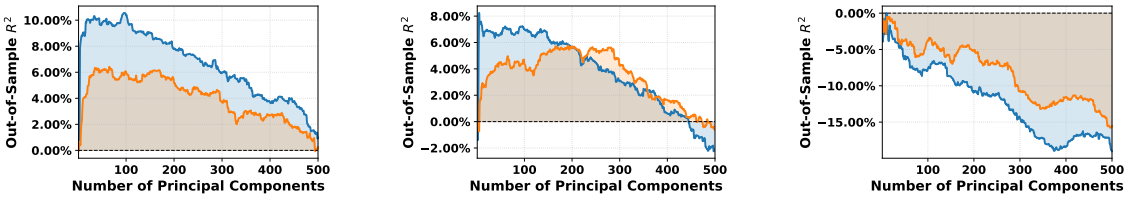
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.8: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Outlook news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



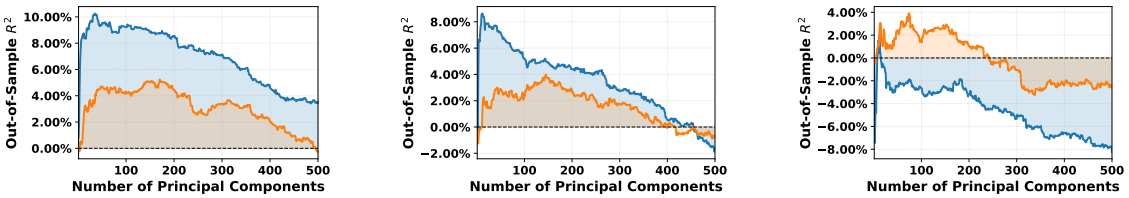
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.9: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Earnings news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



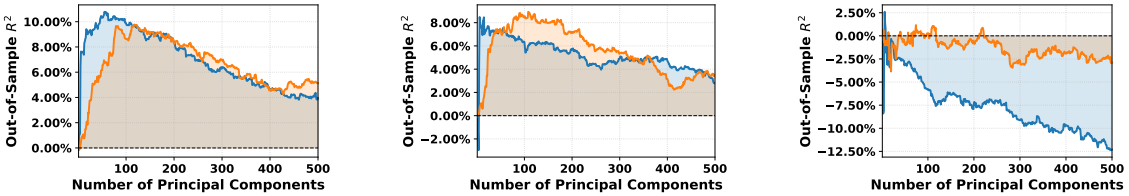
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.10: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Competition news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



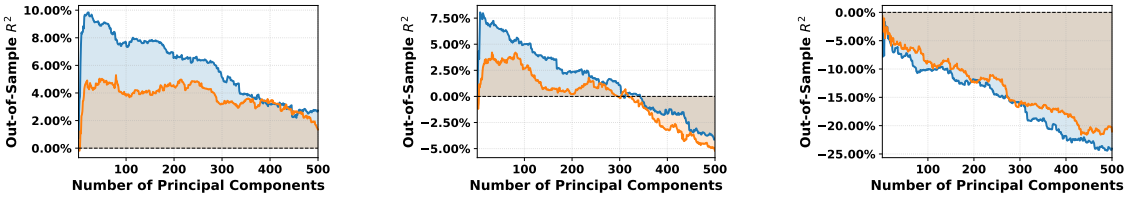
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.11: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Operations news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



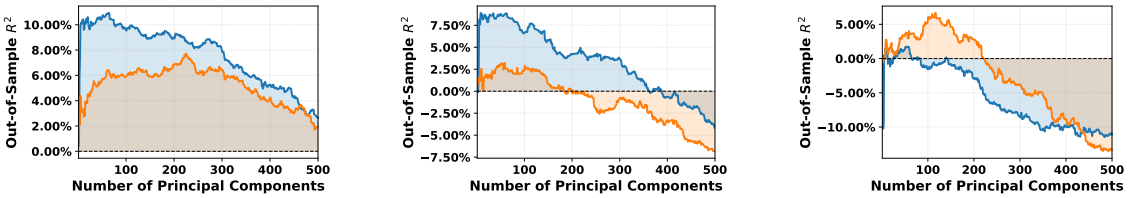
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.12: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Momentum news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



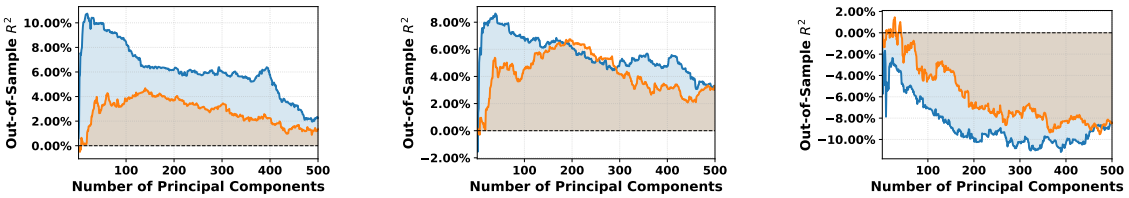
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.13: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Launches news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



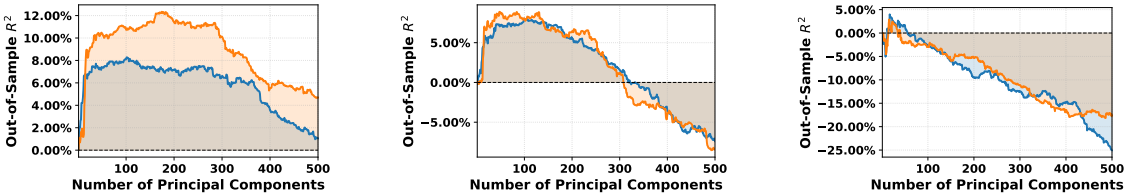
(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.14: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Legal and Regulatory news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.15: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Analyst Ratings news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.



(a) 1-day horizon (b) 5-day horizon (c) 21-day horizon

Figure C.16: Out-of-sample R^2 profiles for "raw news" and "pure news" embeddings for Macro news. Panels (a)–(c) correspond to the 1-day, 5-day, and 21-day forecasting horizons, respectively.

Appendix D

Additional Tests

D.1 Robustness Check

The correlation results reported in Table 4.3 are computed by pooling balanced accuracy–AUC pairs across news source types and news tags. However, the number of articles associated with each news source type or news tag is not constant across categories. For example, the Financial Media source type may be associated with a substantially larger number of articles than the Research source type. A larger number of articles could, in principle, make firm identity easier to detect for that category and could simultaneously affect the corresponding AUC, independently of the extent to which the embedding encodes genuine firm-specific information. This raises the possibility that the observed correlation between balanced accuracy and AUC is, at least in part, confounded by differences in article volume across news source types and news tags, rather than solely reflecting an association between firm identity and predictive performance.

To assess the extent to which this confounding factor affects the results reported in Table 4.3, the correlation between balanced accuracy and AUC is computed, controlling for the number of articles associated with each news source type or news tag. We consider two article count measures. One, the total number of articles, and two, the number of articles in the test window only. Two article count measures are considered because they proxy for two distinct sources of potential confounding. The total article count, combining the training and test windows, proxies for the overall volume of news associated with a given source type or tag, addressing the concern that some categories may simply constitute richer or more heavily covered segments of the news universe, independently of any relationship between firm identity and predictability. On the other hand, test-window article count addresses a narrower concern related to estimation precision. Since both balanced accuracy and AUC are computed out-of-sample, categories with fewer test observations may yield noisier estimates of both quantities, which could itself induce spurious correlation between the two metrics. Besides, considering both measures provides stronger evidence for the robustness of the association between balanced accuracy and AUC.

In addition, an indicator variable denoting whether the embedding is a “raw news” or “pure news” representation is included as a further control, since article counts do not vary between the two representations for a given news source type or news tag.

For each forecasting horizon, balanced accuracy and AUC are separately regressed on the article count measure and the embedding-type indicator, and the resulting residuals are retained. The Pearson correlation coefficient is then computed between the two sets of residuals. This procedure yields the correlation between balanced accuracy and AUC after removing the variation in each that can be explained by article volume and embedding type, and is therefore referred to as the partial correlation.

The resulting partial correlation coefficients and corresponding p-values are reported

in Table D.1.

Table D.1: Partial correlation between balanced accuracy and AUC, controlling for article volume and embedding type.

Horizon	Control Variable	Partial Correlation	p-value
1-day	Total Article Count	0.4777	0.0057
5-day	Total Article Count	0.3346	0.0612
21-day	Total Article Count	0.0761	0.6790
1-day	Test Article Count	0.4341	0.0131
5-day	Test Article Count	0.2813	0.1189
21-day	Test Article Count	0.0431	0.8149

Using a significance level of $\alpha = 0.05$, the positive association between balanced accuracy and AUC at the 1-day horizon remains statistically significant after controlling for article volume, under both the total article count specification ($r = 0.4777$, $p = 0.0057$) and the test article count specification ($r = 0.4341$, $p = 0.0131$), although the magnitude of the correlation is reduced relative to the results in Table 4.3. At the 5-day horizon, the partial correlation is lower than the correlation in Table 4.3 and is not statistically significant at the 5% level under either specification ($p = 0.0612$ and $p = 0.1189$, respectively). At the 21-day horizon, the partial correlation is close to zero and not statistically significant under either specification ($p = 0.6790$ and $p = 0.8149$).

These results suggest that the positive association between firm identifiability and predictive performance at the 1-day horizon is not solely attributable to differences in article volume across news source types and news tags. At the 5-day and 21-day horizons, however, the association is substantially weaker once article volume is controlled for, indicating that a meaningful part of the unadjusted correlation at these horizons may be explained by differences in the number of articles associated with each category, rather than by firm-specific information encoded in the embeddings. As with the analysis in Section 4.6.4, these results should not be interpreted as establishing a causal relationship between firm identity and predictive performance.

Bibliography

[1] Junyu Chen, Tom Boot, Lingwei Kong, and Weining Wang. Transformer-based covar: Systemic risk in textual information. *arXiv preprint arXiv:2602.12490*, 2026.

[2] Yifei Chen, Bryan T Kelly, and Dacheng Xiu. Expected returns and large language models. *Available at SSRN 4416687*, 2022.

[3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, 2019.

[4] Antoine Didisheim, Bryan T Kelly, Mohammad Pourmohammadi, and Hanqing Tian. The inefficient pricing of news. Technical report, National Bureau of Economic Research, 2026.

[5] Nanyi Fei, Yizhao Gao, Zhiwu Lu, and Tao Xiang. Z-score normalization, hubness, and few-shot learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 142–151, 2021.

[6] Joachim Freyberger, Andreas Neuhierl, and Michael Weber. Dissecting characteristics nonparametrically. *The Review of Financial Studies*, 33(5):2326–2377, 2020.

[7] Theis Ingerslev Jensen, Bryan Kelly, and Lasse Heje Pedersen. Is there a replication crisis in finance? *The Journal of Finance*, 78(5):2465–2518, 2023.

[8] JKP Factors. Global factor data. `https://www.jkpfactors.com/`, 2025. Accessed: 15 July 2026.

[9] Daniel Jurafsky and James H. Martin. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3 edition, 2026. Online manuscript released January 6, 2026.

[10] Arman Khaledian, Amirreza Ghadiridehkordi, and Nariman Khaledian. Pca-rag: Principal component analysis for efficient retrieval-augmented generation. *arXiv preprint arXiv:2504.08386*, 2025.

[11] Yaxuan Kong, Yuqi Nie, Xiaowen Dong, John M Mulvey, H Vincent Poor, Qingsong Wen, and Stefan Zohren. Large language models for financial and investment management: Applications and benchmarks. *Journal of Portfolio Management*, 51(2), 2024.

[12] Gang Li, Dandan Qiao, and Mingxuan Zheng. Structured event representation and stock return predictability. *arXiv preprint arXiv:2512.19484*, 2025.

[13] A Craig MacKinlay. Event studies in economics and finance. *Journal of economic literature*, 35(1):13–39, 1997.

[14] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, Cambridge, 2008.

[15] Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*, 2013.

[16] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems*, 26, 2013.

[17] Tomáš Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 conference of the north american chapter of the association for computational linguistics: Human language technologies*, pages 746–751, 2013.

[18] Niklas Muennighoff, Nouamane Tazi, LoÃc Magne, and Nils Reimers. Mteb: Massive text embedding benchmark. *arXiv preprint arXiv:2210.07316*, 2022.

[19] OpenAI. New embedding models and api updates. `https://openai.com/index/new-embedding-models-and-api-updates/`, January 2024. Accessed: 2026-07-12.

[20] Alec Radford, Karthik Narasimhan, Tim Salimans, Ilya Sutskever, et al. Improving language understanding by generative pre-training. 2018.

[21] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 3982–3992, 2019.

[22] Thomas Renault. Sentiment analysis and machine learning in finance: a comparison of methods and models on one million messages. *Digital Finance*, 2(1):1–13, 2020.

[23] Simply Wall St. Largest U.S. stocks by market cap. `https://simplywall.st/stocks/us/market-cap-large`, 2026. Data as of 13 February 2026. Accessed: 21 August 2026.

[24] Sahar Sohangir, Nicholas Petty, and Dingding Wang. Financial sentiment lexicon analysis. In *2018 IEEE 12th international conference on semantic computing (ICSC)*, pages 286–289. IEEE, 2018.

[25] Di-Jia Su, John M Mulvey, and H Vincent Poor. Improving portfolio performance via natural language processing methods. *Journal of Financial Data Science*, 4(2), 2022.

[26] Jianlin Su, Jiarun Cao, Weijie Liu, and Yangyiwen Ou. Whitening sentence representations for better semantics and faster retrieval.(2021). *arXiv preprint arXiv:2103.15316*, 2021.

[27] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.

[28] Mehdi Yekrangi and Neda Abdolvand. Financial markets sentiment analysis: Developing a specialized lexicon. *Journal of Intelligent Information Systems*, 57(1):127–146, 2021.